

where  $b_3$ ,  $b_2$ , and  $b_1$  are determined as follows. By multiplying both sides of this last equation by  $(s + 1)^3$ , we have

$$(s + 1)^3 \frac{B(s)}{A(s)} = b_1(s + 1)^2 + b_2(s + 1) + b_3 \quad (2-16)$$

Then letting  $s = -1$ , Equation (2-16) gives

$$\left[ (s + 1)^3 \frac{B(s)}{A(s)} \right]_{s=-1} = b_3$$

Also, differentiation of both sides of Equation (2-16) with respect to  $s$  yields

$$\frac{d}{ds} \left[ (s + 1)^3 \frac{B(s)}{A(s)} \right] = b_2 + 2b_1(s + 1) \quad (2-17)$$

If we let  $s = -1$  in equation (2-17), then

$$\frac{d}{ds} \left[ (s + 1)^3 \frac{B(s)}{A(s)} \right]_{s=-1} = b_2$$

By differentiating both sides of equation (2-17) with respect to  $s$ , the result is

$$\frac{d^2}{ds^2} \left[ (s + 1)^3 \frac{B(s)}{A(s)} \right] = 2b_1$$

From the preceding analysis it can be seen that the values of  $b_3$ ,  $b_2$ , and  $b_1$  are found systematically as follows:

$$\begin{aligned} b_3 &= \left[ (s + 1)^3 \frac{B(s)}{A(s)} \right]_{s=-1} \\ &= (s^2 + 2s + 3)_{s=-1} \\ &= 2 \\ b_2 &= \left\{ \frac{d}{ds} \left[ (s + 1)^3 \frac{B(s)}{A(s)} \right] \right\}_{s=-1} \\ &= \left[ \frac{d}{ds} (s^2 + 2s + 3) \right]_{s=-1} \\ &= (2s + 2)_{s=-1} \\ &= 0 \\ b_1 &= \frac{1}{2!} \left\{ \frac{d^2}{ds^2} \left[ (s + 1)^3 \frac{B(s)}{A(s)} \right] \right\}_{s=-1} \\ &= \frac{1}{2!} \left[ \frac{d^2}{ds^2} (s^2 + 2s + 3) \right]_{s=-1} \\ &= \frac{1}{2} (2) = 1 \end{aligned}$$

We thus obtain

$$\begin{aligned}
 f(t) &= \mathcal{L}^{-1}[F(s)] \\
 &= \mathcal{L}^{-1}\left[\frac{1}{s+1}\right] + \mathcal{L}^{-1}\left[\frac{0}{(s+1)^2}\right] + \mathcal{L}^{-1}\left[\frac{2}{(s+1)^3}\right] \\
 &= e^{-t} + 0 + t^2 e^{-t} \\
 &= (1 + t^2)e^{-t}, \quad \text{for } t \geq 0
 \end{aligned}$$

**Comments.** For complicated functions with denominators involving higher-order polynomials, partial-fraction expansion may be quite time consuming. In such a case, use of MATLAB is recommended. (See Section 2–6.)

## 2-6 PARTIAL-FRACTION EXPANSION WITH MATLAB

MATLAB has a command to obtain the partial-fraction expansion of  $B(s)/A(s)$ .

Consider the transfer function

$$\frac{B(s)}{A(s)} = \frac{\text{num}}{\text{den}} = \frac{b_0 s^n + b_1 s^{n-1} + \cdots + b_n}{s^n + a_1 s^{n-1} + \cdots + a_n}$$

where some of  $a_i$  and  $b_j$  may be zero. In MATLAB row vectors **num** and **den** specify the coefficients of the numerator and denominator of the transfer function. That is,

$$\begin{aligned}
 \text{num} &= [b_0 \quad b_1 \quad \cdots \quad b_n] \\
 \text{den} &= [1 \quad a_1 \quad \cdots \quad a_n]
 \end{aligned}$$

The command

$$[r,p,k] = \text{residue}(\text{num},\text{den})$$

finds the residues, poles, and direct terms of a partial-fraction expansion of the ratio of two polynomials  $B(s)$  and  $A(s)$ .

The partial-fraction expansion of  $B(s)/A(s)$  is given by

$$\frac{B(s)}{A(s)} = \frac{r(1)}{s - p(1)} + \frac{r(2)}{s - p(2)} + \cdots + \frac{r(n)}{s - p(n)} + k(s) \quad (2-18)$$

Comparing Equations (2-14) and (2-18), we note that  $p(1) = -p_1$ ,  $p(2) = -p_2, \dots$ ,  $p(n) = -p_n$ ;  $r(1) = a_1$ ,  $r(2) = a_2, \dots$ ,  $r(n) = a_n$ . [ $k(s)$  is a direct term.]

### EXAMPLE 2-6

Consider the following transfer function:

$$\frac{B(s)}{A(s)} = \frac{2s^3 + 5s^2 + 3s + 6}{s^3 + 6s^2 + 11s + 6}$$

For this function.

```
num = [2  5  3  6]
den = [1  6 11  6]
```

The command

```
[r,p,k] = residue(num,den)
```

gives the following result:

```
[r,p,k] = residue(num,den)

r =

   -6.0000
   -4.0000
    3.0000

p =

   -3.0000
   -2.0000
   -1.0000

k =

     2
```

(Note that the residues are returned in column vector  $r$ , the pole locations in column vector  $p$ , and the direct term in row vector  $k$ .) This is the MATLAB representation of the following partial-fraction expansion of  $B(s)/A(s)$ ;

$$\begin{aligned}\frac{B(s)}{A(s)} &= \frac{2s^3 + 5s^2 + 3s + 6}{s^3 + 6s^2 + 11s + 6} \\ &= \frac{-6}{s+3} + \frac{-4}{s+2} + \frac{3}{s+1} + 2\end{aligned}$$

The command

```
[num,den] = residue(r,p,k)
```

where  $r$ ,  $p$ , and  $k$  are as given in the previous MATLAB output, converts the partial-fraction expansion back to the polynomial ratio  $B(s)/A(s)$ , as follows:

```
[num,den] = residue(r,p,k)

num =

    2.0000    5.0000    3.0000    6.0000

den =

    1.0000    6.0000   11.0000    6.0000
```

Note that if  $p(j) = p(j+1) = \dots = p(j+m-1)$  [that is,  $p_j = p_{j+1} = \dots = p_{j+m-1}$ ], the pole  $p(j)$  is a pole of multiplicity  $m$ . In such a case, the expansion includes terms of the form

$$\frac{r(j)}{s - p(j)} + \frac{r(j+1)}{[s - p(j)]^2} + \dots + \frac{r(j+m-1)}{[s - p(j)]^m}$$

For details, see Example 2-7.

**EXAMPLE 2-7** Expand the following  $B(s)/A(s)$  into partial-fractions with MATLAB.

$$\frac{B(s)}{A(s)} = \frac{s^2 + 2s + 3}{(s + 1)^3} = \frac{s^2 + 2s + 3}{s^3 + 3s^2 + 3s + 1}$$

For this function, we have

```
num = [0  1  2  3]
den = [1  3  3  1]
```

The command

```
[r,p,k] = residue(num,den)
```

gives the result shown on the next page. It is the MATLAB representation of the following partial-fraction expansion of  $B(s)/A(s)$ :

$$\frac{B(s)}{A(s)} = \frac{1}{s + 1} + \frac{0}{(s + 1)^2} + \frac{2}{(s + 1)^3}$$

Note that the direct term  $k$  is zero.

```
num = [0 1 2 3];  
den = [1 3 3 1];  
[r,p,k] = residue(num,den)
```

```
r =
```

```
1.0000  
0.0000  
2.0000
```

```
p =
```

```
-1.0000  
-1.0000  
-1.0000
```

```
k =
```

```
[]
```

## 2-7 SOLVING LINEAR, TIME-INVARIANT, DIFFERENTIAL EQUATIONS

In this section we are concerned with the use of the Laplace transform method in solving linear, time-invariant, differential equations.

The Laplace transform method yields the complete solution (complementary solution and particular solution) of linear, time-invariant, differential equations. Classical methods for finding the complete solution of a differential equation require the evaluation of the integration constants from the initial conditions. In the case of the Laplace transform method, however, this requirement is unnecessary because the initial conditions are automatically included in the Laplace transform of the differential equation.

If all initial conditions are zero, then the Laplace transform of the differential equation is obtained simply by replacing  $d/dt$  with  $s$ ,  $d^2/dt^2$  with  $s^2$ , and so on.

In solving linear, time-invariant, differential equations by the Laplace transform method, two steps are involved.

1. By taking the Laplace transform of each term in the given differential equation, convert the differential equation into an algebraic equation in  $s$  and obtain the

expression for the Laplace transform of the dependent variable by rearranging the algebraic equation.

2. The time solution of the differential equation is obtained by finding the inverse Laplace transform of the dependent variable.

In the following discussion, two examples are used to demonstrate the solution of linear, time-invariant, differential equations by the Laplace transform method.

### EXAMPLE 2-8

Find the solution  $x(t)$  of the differential equation

$$\ddot{x} + 3\dot{x} + 2x = 0, \quad x(0) = a, \quad \dot{x}(0) = b$$

where  $a$  and  $b$  are constants.

By writing the Laplace transform of  $x(t)$  as  $X(s)$  or

$$\mathcal{L}[x(t)] = X(s)$$

we obtain

$$\mathcal{L}[\dot{x}] = sX(s) - x(0)$$

$$\mathcal{L}[\ddot{x}] = s^2X(s) - sx(0) - \dot{x}(0)$$

And so the given differential equation becomes

$$[s^2X(s) - sx(0) - \dot{x}(0)] + 3[sX(s) - x(0)] + 2X(s) = 0$$

By substituting the given initial conditions into this last equation, we obtain

$$[s^2X(s) - as - b] + 3[sX(s) - a] + 2X(s) = 0$$

or

$$(s^2 + 3s + 2)X(s) = as + b + 3a$$

Solving for  $X(s)$ , we have

$$X(s) = \frac{as + b + 3a}{s^2 + 3s + 2} = \frac{as + b + 3a}{(s + 1)(s + 2)} = \frac{2a + b}{s + 1} - \frac{a + b}{s + 2}$$

The inverse Laplace transform of  $X(s)$  gives

$$\begin{aligned} x(t) &= \mathcal{L}^{-1}[X(s)] = \mathcal{L}^{-1}\left[\frac{2a + b}{s + 1}\right] - \mathcal{L}^{-1}\left[\frac{a + b}{s + 2}\right] \\ &= (2a + b)e^{-t} - (a + b)e^{-2t}, \quad \text{for } t \geq 0 \end{aligned}$$

which is the solution of the given differential equation. Notice that the initial conditions  $a$  and  $b$  appear in the solution. Thus  $x(t)$  has no undetermined constants.

### EXAMPLE 2-9

Find the solution  $x(t)$  of the differential equation

$$\ddot{x} + 2\dot{x} + 5x = 3, \quad x(0) = 0, \quad \dot{x}(0) = 0$$

Noting that  $\mathcal{L}[3] = 3/s$ ,  $x(0) = 0$ , and  $\dot{x}(0) = 0$ , the Laplace transform of the differential equation becomes

$$s^2X(s) + 2sX(s) + 5X(s) = \frac{3}{s}$$

Solving for  $X(s)$ , we find

$$\begin{aligned} X(s) &= \frac{3}{s(s^2 + 2s + 5)} = \frac{3}{5} \frac{1}{s} - \frac{3}{5} \frac{s + 2}{s^2 + 2s + 5} \\ &= \frac{3}{5} \frac{1}{s} - \frac{3}{10} \frac{2}{(s + 1)^2 + 2^2} - \frac{3}{5} \frac{s + 1}{(s + 1)^2 + 2^2} \end{aligned}$$

Hence the inverse Laplace transform becomes

$$\begin{aligned} x(t) &= \mathcal{L}^{-1}[X(s)] \\ &= \frac{3}{5} \mathcal{L}^{-1}\left[\frac{1}{s}\right] - \frac{3}{10} \mathcal{L}^{-1}\left[\frac{2}{(s + 1)^2 + 2^2}\right] - \frac{3}{5} \mathcal{L}^{-1}\left[\frac{s + 1}{(s + 1)^2 + 2^2}\right] \\ &= \frac{3}{5} - \frac{3}{10} e^{-t} \sin 2t - \frac{3}{5} e^{-t} \cos 2t, \quad \text{for } t \geq 0 \end{aligned}$$

which is the solution of the given differential equation.

## EXAMPLE PROBLEMS AND SOLUTIONS

**A-2-1.** Find the poles of the following  $F(s)$ :

$$F(s) = \frac{1}{1 - e^{-s}}$$

**Solution.** The poles are found from

$$e^{-s} = 1$$

or

$$e^{-(\sigma + j\omega)} = e^{-\sigma}(\cos \omega - j \sin \omega) = 1$$

From this it follows that  $\sigma = 0$ ,  $\omega = \pm 2n\pi$  ( $n = 0, 1, 2, \dots$ ). Thus, the poles are located at

$$s = \pm j2n\pi \quad (n = 0, 1, 2, \dots)$$

**A-2-2.** Find the Laplace transform of  $f(t)$  defined by

$$\begin{aligned} f(t) &= 0, & \text{for } t < 0 \\ &= te^{-3t}, & \text{for } t \geq 0 \end{aligned}$$

**Solution.** Since

$$\mathcal{L}[t] = G(s) = \frac{1}{s^2}$$

referring to Equation (2-6), we obtain

$$F(s) = \mathcal{L}[te^{-3t}] = G(s + 3) = \frac{1}{(s + 3)^2}$$

**A-2-3.** What is the Laplace transform of

$$\begin{aligned} f(t) &= 0, & \text{for } t < 0 \\ &= \sin(\omega t + \theta), & \text{for } t \geq 0 \end{aligned}$$

where  $\theta$  is a constant?

**Solution.** Noting that

$$\sin(\omega t + \theta) = \sin \omega t \cos \theta + \cos \omega t \sin \theta$$

we have

$$\begin{aligned}\mathcal{L}[\sin(\omega t + \theta)] &= \cos \theta \mathcal{L}[\sin \omega t] + \sin \theta \mathcal{L}[\cos \omega t] \\ &= \cos \theta \frac{\omega}{s^2 + \omega^2} + \sin \theta \frac{s}{s^2 + \omega^2} \\ &= \frac{\omega \cos \theta + s \sin \theta}{s^2 + \omega^2}\end{aligned}$$

**A-2-4.** Find the Laplace transform  $F(s)$  of the function  $f(t)$  shown in Figure 2-3. Also find the limiting value of  $F(s)$  as  $a$  approaches zero.

**Solution.** The function  $f(t)$  can be written

$$f(t) = \frac{1}{a^2} 1(t) - \frac{2}{a^2} 1(t - a) + \frac{1}{a^2} 1(t - 2a)$$

Then

$$\begin{aligned}F(s) &= \mathcal{L}[f(t)] \\ &= \frac{1}{a^2} \mathcal{L}[1(t)] - \frac{2}{a^2} \mathcal{L}[1(t - a)] + \frac{1}{a^2} \mathcal{L}[1(t - 2a)] \\ &= \frac{1}{a^2} \frac{1}{s} - \frac{2}{a^2} \frac{1}{s} e^{-as} + \frac{1}{a^2} \frac{1}{s} e^{-2as} \\ &= \frac{1}{a^2 s} (1 - 2e^{-as} + e^{-2as})\end{aligned}$$

As  $a$  approaches zero, we have

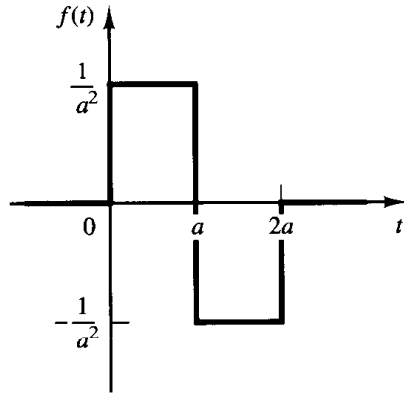
$$\begin{aligned}\lim_{a \rightarrow 0} F(s) &= \lim_{a \rightarrow 0} \frac{1 - 2e^{-as} + e^{-2as}}{a^2 s} = \lim_{a \rightarrow 0} \frac{\frac{d}{da} (1 - 2e^{-as} + e^{-2as})}{\frac{d}{da} (a^2 s)} \\ &= \lim_{a \rightarrow 0} \frac{2se^{-as} - 2se^{-2as}}{2as} = \lim_{a \rightarrow 0} \frac{e^{-as} - e^{-2as}}{a} \\ &= \lim_{a \rightarrow 0} \frac{\frac{d}{da} (e^{-as} - e^{-2as})}{\frac{d}{da} (a)} = \lim_{a \rightarrow 0} \frac{-se^{-as} + 2se^{-2as}}{1} \\ &= -s + 2s = s\end{aligned}$$

**A-2-5.** Find the initial value of  $df(t)/dt$  when the Laplace transform of  $f(t)$  is given by

$$F(s) = \mathcal{L}[f(t)] = \frac{2s + 1}{s^2 + s + 1}$$

**Solution.** Using the initial-value theorem,





**Figure 2-3**  
Function  $f(t)$ .

$$\lim_{t \rightarrow 0+} f(t) = f(0+) = \lim_{s \rightarrow \infty} sF(s) = \lim_{s \rightarrow \infty} \frac{s(2s+1)}{s^2+s+1} = 2$$

Since the  $\mathcal{L}_+$  transform of  $df(t)/dt = g(t)$  is given by

$$\begin{aligned} \mathcal{L}_+[g(t)] &= sF(s) - f(0+) \\ &= \frac{s(2s+1)}{s^2+s+1} - 2 = \frac{-s-2}{s^2+s+1} \end{aligned}$$

the initial value of  $df(t)/dt$  is obtained as

$$\begin{aligned} \lim_{t \rightarrow 0+} \frac{df(t)}{dt} &= g(0+) = \lim_{s \rightarrow \infty} s[sF(s) - f(0+)] \\ &= \lim_{s \rightarrow \infty} \frac{-s^2 - 2s}{s^2 + s + 1} = -1 \end{aligned}$$

**A-2-6.** The derivative of the unit-impulse function  $\delta(t)$  is called a *unit-doublet* function. (Thus, the integral of the unit-doublet function is the unit-impulse function.) Mathematically, an example of the unit-doublet function, which is usually denoted by  $u_2(t)$ , may be given by

$$u_2(t) = \lim_{t_0 \rightarrow 0} \frac{1(t) - 2[1(t - t_0)] + 1(t - 2t_0)}{t_0^2}$$

Obtain the Laplace transform of  $u_2(t)$ .

**Solution.** The Laplace transform of  $u_2(t)$  is given by

$$\begin{aligned} \mathcal{L}[u_2(t)] &= \lim_{t_0 \rightarrow 0} \frac{1}{t_0^2} \left( \frac{1}{s} - \frac{2}{s} e^{-t_0 s} + \frac{1}{s} e^{-2t_0 s} \right) \\ &= \lim_{t_0 \rightarrow 0} \frac{1}{t_0^2 s} \left[ 1 - 2 \left( 1 - t_0 s + \frac{t_0^2 s^2}{2} + \cdots \right) + \left( 1 - 2t_0 s + \frac{4t_0^2 s^2}{2} + \cdots \right) \right] \\ &= \lim_{t_0 \rightarrow 0} \frac{1}{t_0^2 s} [t_0^2 s^2 + (\text{higher-order terms in } t_0 s)] = s \end{aligned}$$

**A-2-7.** Find the Laplace transform of  $f(t)$  defined by

$$\begin{aligned} f(t) &= 0, & \text{for } t < 0 \\ &= t^2 \sin \omega t, & \text{for } t \geq 0 \end{aligned}$$

**Solution.** Since

$$\mathcal{L}[\sin \omega t] = \frac{\omega}{s^2 + \omega^2}$$

applying the complex-differentiation theorem

$$\mathcal{L}[t^2 f(t)] = \frac{d^2}{ds^2} F(s)$$

to this problem, we have

$$\mathcal{L}[f(t)] = \mathcal{L}[t^2 \sin \omega t] = \frac{d^2}{ds^2} \left[ \frac{\omega}{s^2 + \omega^2} \right] = \frac{-2\omega^3 + 6\omega s^2}{(s^2 + \omega^2)^3}$$

**A-2-8.** Prove that if  $f(t)$  is of exponential order and if  $\int_0^\infty f(t) dt$  exists [which means that  $\int_0^\infty f(t) dt$  assumes a definite value] then

$$\int_0^\infty f(t) dt = \lim_{s \rightarrow 0} F(s)$$

where  $F(s) = \mathcal{L}[f(t)]$ .

**Solution.** Note that

$$\int_0^\infty f(t) dt = \lim_{t \rightarrow \infty} \int_0^t f(t) dt$$

Referring to equation (2-9),

$$\mathcal{L} \left[ \int_0^t f(t) dt \right] = \frac{F(s)}{s}$$

Since  $\int_0^\infty f(t) dt$  exists, by applying the final-value theorem to this case,

$$\lim_{t \rightarrow \infty} \int_0^t f(t) dt = \lim_{s \rightarrow 0} s \frac{F(s)}{s}$$

or

$$\int_0^\infty f(t) dt = \lim_{s \rightarrow 0} F(s)$$

**A-2-9.** Determine the Laplace transform of the convolution integral

$$f_1(t) * f_2(t) = \int_0^t \tau [1 - e^{-(t-\tau)}] d\tau = \int_0^t (t - \tau)(1 - e^{-\tau}) d\tau$$

where

$$\begin{aligned} f_1(t) &= f_2(t) = 0, & \text{for } t < 0 \\ f_1(t) &= t, & \text{for } t \geq 0 \\ f_2(t) &= 1 - e^{-t}, & \text{for } t \geq 0 \end{aligned}$$

**Solution.** Note that

$$\mathcal{L}[t] = F_1(s) = \frac{1}{s^2}$$

$$\mathcal{L}[1 - e^{-t}] = F_2(s) = \frac{1}{s} - \frac{1}{s+1}$$

The Laplace transform of the convolution integral is given by

$$\begin{aligned}\mathcal{L}[f_1(t)*f_2(t)] &= F_1(s)F_2(s) = \frac{1}{s^2} \left( \frac{1}{s} - \frac{1}{s+1} \right) \\ &= \frac{1}{s^3} - \frac{1}{s^2(s+1)} = \frac{1}{s^3} - \frac{1}{s^2} + \frac{1}{s} - \frac{1}{s+1}\end{aligned}$$

To verify that it is indeed the Laplace transform of the convolution integral, let us first perform integration of the convolution integral and then take its Laplace transform.

$$\begin{aligned}f_1(t)*f_2(t) &= \int_0^t \tau [1 - e^{-(t-\tau)}] d\tau = \int_0^t (t-\tau)(1 - e^{-\tau}) d\tau \\ &= \frac{t^2}{2} - t + 1 - e^{-t}\end{aligned}$$

And so

$$\mathcal{L}\left[\frac{t^2}{2} - t + 1 - e^{-t}\right] = \frac{1}{s^3} - \frac{1}{s^2} + \frac{1}{s} - \frac{1}{s+1}$$

**A-2-10.** Prove that if  $f(t)$  is a periodic function with period  $T$  then

$$\mathcal{L}[f(t)] = \frac{\int_0^T f(t)e^{-st} dt}{1 - e^{-Ts}}$$

**Solution.**

$$\mathcal{L}[f(t)] = \int_0^\infty f(t)e^{-st} dt = \sum_{n=0}^\infty \int_{nT}^{(n+1)T} f(t)e^{-st} dt$$

By changing the independent variable from  $t$  to  $\tau$ , where  $\tau = t - nT$ ,

$$\mathcal{L}[f(t)] = \sum_{n=0}^\infty e^{-nTs} \int_0^T f(\tau)e^{-s\tau} d\tau$$

Noting that

$$\begin{aligned}\sum_{n=0}^\infty e^{-nTs} &= 1 + e^{-Ts} + e^{-2Ts} + \dots \\ &= 1 + e^{-Ts}(1 + e^{-Ts} + e^{-2Ts} + \dots) \\ &= 1 + e^{-Ts} \left( \sum_{n=0}^\infty e^{-nTs} \right)\end{aligned}$$

we obtain

$$\sum_{n=0}^\infty e^{-nTs} = \frac{1}{1 - e^{-Ts}}$$

It follows that

$$\mathcal{L}[f(t)] = \frac{\int_0^T f(t)e^{-st} dt}{1 - e^{-Ts}}$$

**A-2-11.** What is the Laplace transform of the periodic function shown in Figure 2-4?

**Solution.** Note that

$$\begin{aligned}\int_0^T f(t)e^{-st} dt &= \int_0^{T/2} e^{-st} dt + \int_{T/2}^T (-1)e^{-st} dt \\ &= \frac{e^{-st}}{-s} \Big|_0^{T/2} - \frac{e^{-st}}{-s} \Big|_{T/2}^T \\ &= \frac{e^{-(1/2)Ts} - 1}{-s} + \frac{e^{-Ts} - e^{-(1/2)Ts}}{s} \\ &= \frac{1}{s} [e^{-Ts} - 2e^{-(1/2)Ts} + 1] \\ &= \frac{1}{s} [1 - e^{-(1/2)Ts}]^2\end{aligned}$$

Referring to Problem A-2-10, we have

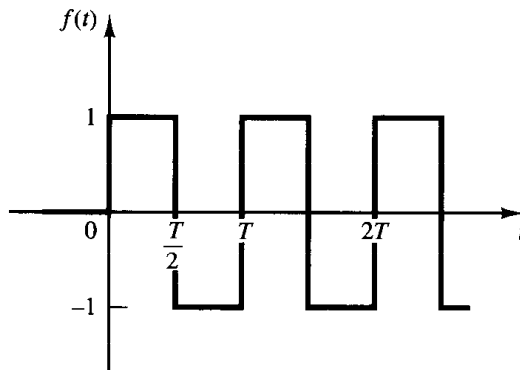
$$\begin{aligned}F(s) &= \frac{\int_0^T f(t)e^{-st} dt}{1 - e^{-Ts}} = \frac{(1/s)[1 - e^{-(1/2)Ts}]^2}{1 - e^{-Ts}} \\ &= \frac{1 - e^{-(1/2)Ts}}{s[1 + e^{-(1/2)Ts}]} = \frac{1}{s} \tanh \frac{Ts}{4}\end{aligned}$$

**A-2-12.** Find the inverse Laplace transform of  $F(s)$ , where

$$F(s) = \frac{1}{s(s^2 + 2s + 2)}$$

**Solution.** Since

$$s^2 + 2s + 2 = (s + 1 + j1)(s + 1 - j1)$$



**Figure 2-4**  
Periodic function (square wave).

we notice that  $F(s)$  involves a pair of complex-conjugate poles, and so we expand  $F(s)$  into the form

$$F(s) = \frac{1}{s(s^2 + 2s + 2)} = \frac{a_1}{s} + \frac{a_2s + a_3}{s^2 + 2s + 2}$$

where  $a_1, a_2$ , and  $a_3$  are determined from

$$1 = a_1(s^2 + 2s + 2) + (a_2s + a_3)s$$

By comparing coefficients of  $s^2$ ,  $s$ , and  $s^0$  terms on both sides of this last equation, respectively, we obtain

$$a_1 + a_2 = 0, \quad 2a_1 + a_3 = 0, \quad 2a_1 = 1$$

from which

$$a_1 = \frac{1}{2}, \quad a_2 = -\frac{1}{2}, \quad a_3 = -1$$

Therefore,

$$\begin{aligned} F(s) &= \frac{1}{2} \frac{1}{s} - \frac{1}{2} \frac{s+2}{s^2 + 2s + 2} \\ &= \frac{1}{2} \frac{1}{s} - \frac{1}{2} \frac{1}{(s+1)^2 + 1^2} - \frac{1}{2} \frac{s+1}{(s+1)^2 + 1^2} \end{aligned}$$

The inverse Laplace transform of  $F(s)$  gives

$$f(t) = \frac{1}{2} - \frac{1}{2} e^{-t} \sin t - \frac{1}{2} e^{-t} \cos t, \quad \text{for } t \geq 0$$

**A-2-13.** Obtain the inverse Laplace transform of

$$F(s) = \frac{5(s+2)}{s^2(s+1)(s+3)}$$

**Solution.**

$$F(s) = \frac{5(s+2)}{s^2(s+1)(s+3)} = \frac{b_1}{s} + \frac{b_2}{s^2} + \frac{a_1}{s+1} + \frac{a_2}{s+3}$$

where

$$\begin{aligned} a_1 &= \left. \frac{5(s+2)}{s^2(s+3)} \right|_{s=-1} = \frac{5}{2} \\ a_2 &= \left. \frac{5(s+2)}{s^2(s+1)} \right|_{s=-3} = \frac{5}{18} \\ b_2 &= \left. \frac{5(s+2)}{(s+1)(s+3)} \right|_{s=0} = \frac{10}{3} \\ b_1 &= \frac{d}{ds} \left[ \frac{5(s+2)}{(s+1)(s+3)} \right]_{s=0} \\ &= \left. \frac{5(s+1)(s+3) - 5(s+2)(2s+4)}{(s+1)^2(s+3)^2} \right|_{s=0} = -\frac{25}{9} \end{aligned}$$

Thus

$$F(s) = -\frac{25}{9} \frac{1}{s} + \frac{10}{3} \frac{1}{s^2} + \frac{5}{2} \frac{1}{s+1} + \frac{5}{18} \frac{1}{s+3}$$

The inverse Laplace transform of  $F(s)$  is

$$f(t) = -\frac{25}{9} + \frac{10}{3}t + \frac{5}{2}e^{-t} + \frac{5}{18}e^{-3t}, \quad \text{for } t \geq 0$$

**A-2-14.** Find the inverse Laplace transform of

$$F(s) = \frac{s^4 + 2s^3 + 3s^2 + 4s + 5}{s(s+1)}$$

**Solution.** Since the numerator polynomial is of higher degree than the denominator polynomial, by dividing the numerator by the denominator until the remainder is a fraction, we obtain

$$F(s) = s^2 + s + 2 + \frac{2s+5}{s(s+1)} = s^2 + s + 2 + \frac{a_1}{s} + \frac{a_2}{s+1}$$

where

$$a_1 = \left. \frac{2s+5}{s+1} \right|_{s=0} = 5$$

$$a_2 = \left. \frac{2s+5}{s} \right|_{s=-1} = -3$$

It follows that

$$F(s) = s^2 + s + 2 + \frac{5}{s} - \frac{3}{s+1}$$

The inverse Laplace transform of  $F(s)$  is

$$f(t) = \mathcal{L}^{-1}[F(s)] = \frac{d^2}{dt^2} \delta(t) + \frac{d}{dt} \delta(t) + 2\delta(t) + 5 - 3e^{-t}, \quad \text{for } t \geq 0$$

**A-2-15.** Derive the inverse Laplace transform of

$$F(s) = \frac{1}{s(s^2 + \omega^2)}$$

**Solution.**

$$F(s) = \frac{1}{s(s^2 + \omega^2)} = \frac{1}{\omega^2} \frac{1}{s} - \frac{1}{\omega^2} \frac{s}{s^2 + \omega^2}$$

Hence the inverse Laplace transform of  $F(s)$  is obtained as

$$f(t) = \mathcal{L}^{-1}[F(s)] = \frac{1}{\omega^2} (1 - \cos \omega t), \quad \text{for } t \geq 0$$

**A-2-16.** Obtain the inverse Laplace transform of the following  $F(s)$ :

$$F(s) = \frac{B(s)}{A(s)} = \frac{B(s)}{(s+p_1)(s+p_{r+1})(s+p_{r+2}) \cdots (s+p_n)}$$

where the degree of polynomial  $B(s)$  is lower than that of polynomial  $A(s)$ .

**Solution.** The partial-fraction expansion of  $F(s)$  is

$$F(s) = \frac{B(s)}{A(s)} = \frac{b_1}{s + p_1} + \frac{b_2}{(s + p_1)^2} + \cdots + \frac{b_{r-1}}{(s + p_1)^{r-1}} + \frac{b_r}{(s + p_1)^r} \\ + \frac{a_{r+1}}{s + p_{r+1}} + \frac{a_{r+2}}{s + p_{r+2}} + \cdots + \frac{a_n}{s + p_n} \quad (2-19)$$

where  $b_n, b_{r-1}, \dots, b_1$  are given by

$$b_r = \left[ (s + p_1)^r \frac{B(s)}{A(s)} \right]_{s=-p_1} \\ b_{r-1} = \left\{ \frac{d}{ds} \left[ (s + p_1)^r \frac{B(s)}{A(s)} \right] \right\}_{s=-p_1} \\ \vdots \\ b_{r-j} = \frac{1}{j!} \left\{ \frac{d^j}{ds^j} \left[ (s + p_1)^r \frac{B(s)}{A(s)} \right] \right\}_{s=-p_1} \\ \vdots \\ b_1 = \frac{1}{(r-1)!} \left\{ \frac{d^{r-1}}{ds^{r-1}} \left[ (s + p_1)^r \frac{B(s)}{A(s)} \right] \right\}_{s=-p_1}$$

The foregoing relationships for the  $b$ 's may be obtained as follows: By multiplying both sides of Equation (2-19) by  $(s + p_1)^r$  and letting  $s$  approach  $-p_1$ , we obtain

$$b_r = \left[ (s + p_1)^r \frac{B(s)}{A(s)} \right]_{s=-p_1}$$

If we multiply both sides of Equation (2-19) by  $(s + p_1)^r$  and then differentiate with respect to  $s$ ,

$$\frac{d}{ds} \left[ (s + p_1)^r \frac{B(s)}{A(s)} \right] = b_r \frac{d}{ds} \left[ \frac{(s + p_1)^r}{(s + p_1)^r} \right] + b_{r-1} \frac{d}{ds} \left[ \frac{(s + p_1)^r}{(s + p_1)^{r-1}} \right] \\ + \cdots + b_1 \frac{d}{ds} \left[ \frac{(s + p_1)^r}{s + p_1} \right] + a_{r+1} \frac{d}{ds} \left[ \frac{(s + p_1)^r}{s + p_{r+1}} \right] \\ + \cdots + a_n \frac{d}{ds} \left[ \frac{(s + p_1)^r}{s + p_n} \right]$$

The first term on the right-hand side of this last equation vanishes. The second term becomes  $b_{r-1}$ . Each of the other terms contains some power of  $(s + p_1)$  as a factor, with the result that, when  $s$  is allowed to approach  $-p_1$ , these terms drop out. Hence

$$b_{r-1} = \lim_{s \rightarrow -p_1} \frac{d}{ds} \left[ (s + p_1)^r \frac{B(s)}{A(s)} \right] \\ = \left\{ \frac{d}{ds} \left[ (s + p_1)^r \frac{B(s)}{A(s)} \right] \right\}_{s=-p_1}$$

Similarly, by performing successive differentiations with respect to  $s$  and by letting  $s$  approach  $-p_1$ , we obtain equations for the  $b_{r-j}$ , where  $j = 2, 3, \dots, r-1$ .

Note that the inverse Laplace transform of  $1/(s + p_1)^n$  is given by

$$\mathcal{L}^{-1}\left[\frac{1}{(s + p_1)^n}\right] = \frac{t^{n-1}}{(n-1)!} e^{-p_1 t}$$

The constants  $a_{r+1}, a_{r+2}, \dots, a_n$  in Equation (2-19) are determined from

$$a_k = \left[ (s + p_k) \frac{B(s)}{A(s)} \right]_{s=-p_k}, \quad \text{for } k = r + 1, r + 2, \dots, n$$

The inverse Laplace transform of  $F(s)$  is then obtained as follows:

$$f(t) = \mathcal{L}^{-1}[F(s)] = \left[ b_1 + b_2 t + \dots + \frac{b_{r-1}}{(r-2)!} t^{r-2} + \frac{b_r}{(r-1)!} t^{r-1} \right] e^{-p_1 t} \\ + a_{r+1} e^{-p_{r+1} t} + a_{r+2} e^{-p_{r+2} t} + \dots + a_n e^{-p_n t}, \quad \text{for } t \geq 0$$

**A-2-17.** Find the Laplace transform of the following differential equation:

$$\ddot{x} + 3\dot{x} + 6x = 0, \quad x(0) = 0, \quad \dot{x}(0) = 3$$

Taking the inverse Laplace transform of  $X(s)$ , obtain the time solution  $x(t)$ .

**Solution.** The Laplace transform of the differential equation is

$$s^2 X(s) - sx(0) - \dot{x}(0) + 3sX(s) - 3x(0) + 6X(s) = 0$$

Substituting the initial conditions and solving for  $X(s)$ ,

$$X(s) = \frac{3}{s^2 + 3s + 6} = \frac{2\sqrt{3}}{\sqrt{5}} \frac{\frac{\sqrt{15}}{2}}{(s + 1.5)^2 + \left(\frac{\sqrt{15}}{2}\right)^2}$$

The inverse Laplace transform of  $X(s)$  is

$$x(t) = \frac{2\sqrt{3}}{\sqrt{5}} e^{-1.5t} \sin\left(\frac{\sqrt{15}}{2} t\right)$$

## PROBLEMS

**B-2-1.** Find the Laplace transforms of the following functions:

$$(a) \quad f_1(t) = 0, \quad \text{for } t < 0 \\ = e^{-0.4t} \cos 12t, \quad \text{for } t \geq 0$$

$$(b) \quad f_2(t) = 0, \quad \text{for } t < 0 \\ = \sin\left(4t + \frac{\pi}{3}\right), \quad \text{for } t \geq 0$$

**B-2-2.** Find the Laplace transforms of the following functions:

$$(a) \quad f_1(t) = 0, \quad \text{for } t < 0 \\ = 3 \sin(5t + 45^\circ) \quad \text{for } t \geq 0 \\ (b) \quad f_2(t) = 0, \quad \text{for } t < 0 \\ = 0.03(1 - \cos 2t) \quad \text{for } t \geq 0$$

**B-2-3.** Obtain the Laplace transform of the function defined by

$$f(t) = 0, \quad \text{for } t < 0 \\ = t^2 e^{-at} \quad \text{for } t \geq 0$$

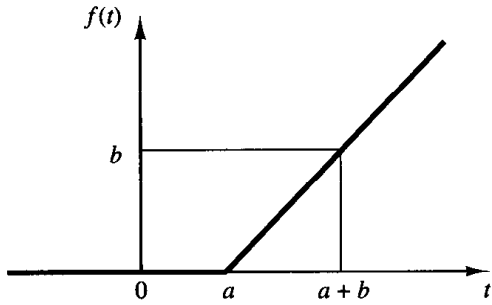


**B-2-4.** Obtain the Laplace transform of the function defined by

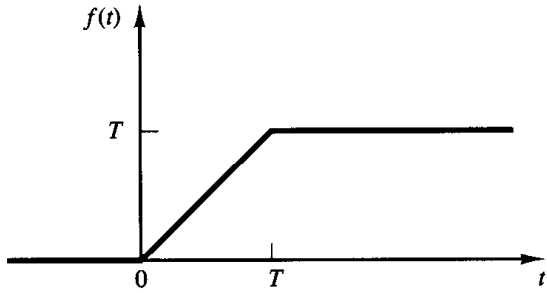
$$\begin{aligned} f(t) &= 0, & \text{for } t < 0 \\ &= \cos 2\omega t \cdot \cos 3\omega t, & \text{for } t \geq 0 \end{aligned}$$

**B-2-5.** What is the Laplace transform of the function  $f(t)$  shown in Figure 2-5?

**B-2-6.** Obtain the Laplace transform of the function  $f(t)$  shown in Figure 2-6.

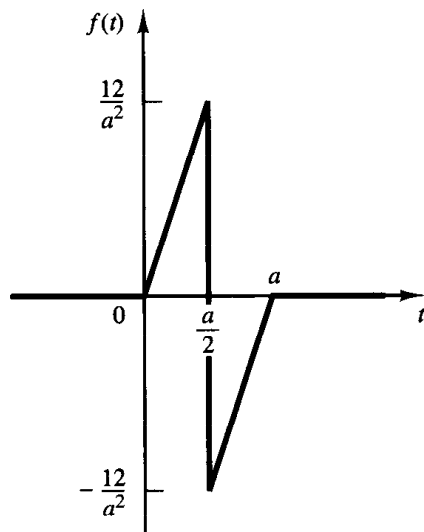


**Figure 2-5**  
Function  $f(t)$ .



**Figure 2-6**

Function  $f(t)$ .



**Figure 2-7**  
Function  $f(t)$ .

**B-2-7.** Find the Laplace transform of the function  $f(t)$  shown in Figure 2-7. Also, find the limiting value of  $\mathcal{L}[f(t)]$  as  $a$  approaches zero.

**B-2-8.** By applying the final-value theorem, find the final value of  $f(t)$  whose Laplace transform is given by

$$F(s) = \frac{10}{s(s+1)}$$

Verify this result by taking the inverse Laplace transform of  $F(s)$  and letting  $t \rightarrow \infty$ .

**B-2-9.** Given

$$F(s) = \frac{1}{(s+2)^2}$$

determine the values of  $f(0+)$  and  $\dot{f}(0+)$ . (Use the initial-value theorem.)

**B-2-10.** Find the inverse Laplace transform of

$$F(s) = \frac{s+1}{s(s^2+s+1)}$$

**B-2-11.** Find the inverse Laplace transforms of the following functions:

(a) 
$$F_1(s) = \frac{6s+3}{s^2}$$

(b) 
$$F_2(s) = \frac{5s+2}{(s+1)(s+2)^2}$$

**B-2-12.** Find the inverse Laplace transform of

$$F(s) = \frac{1}{s^2(s^2+\omega^2)}$$

**B-2-13.** What is the solution of the following differential equation?

$$2\ddot{x} + 7\dot{x} + 3x = 0, \quad x(0) = 3, \quad \dot{x}(0) = 0$$

**B-2-14.** Solve the differential equation

$$\dot{x} + 2x = \delta(t), \quad x(0-) = 0$$

**B-2-15.** Solve the following differential equation:

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x = 0, \quad x(0) = a, \quad \dot{x}(0) = b$$

where  $a$  and  $b$  are constants.

**B-2-16.** Obtain the solution of the differential equation

$$\dot{x} + ax = A \sin \omega t, \quad x(0) = b$$

# 3

## Mathematical Modeling of Dynamic Systems

### 3-1 INTRODUCTION

In studying control systems the reader must be able to model dynamic systems and analyze dynamic characteristics. A mathematical model of a dynamic system is defined as a set of equations that represents the dynamics of the system accurately or, at least, fairly well. Note that a mathematical model is not unique to a given system. A system may be represented in many different ways and, therefore, may have many mathematical models, depending on one's perspective.

The dynamics of many systems, whether they are mechanical, electrical, thermal, economic, biological, and so on, may be described in terms of differential equations. Such differential equations may be obtained by using physical laws governing a particular system, for example, Newton's laws for mechanical systems and Kirchhoff's laws for electrical systems. We must always keep in mind that deriving a reasonable mathematical model is the most important part of the entire analysis.

**Mathematical models.** Mathematical models may assume many different forms. Depending on the particular system and the particular circumstances, one mathematical model may be better suited than other models. For example, in optimal control problems, it is advantageous to use state-space representations. On the other hand, for the transient-response or frequency-response analysis of single-input-single-output, linear, time-invariant systems, the transfer function representation may be more convenient than any other. Once a mathematical model of a system is obtained, various analytical and computer tools can be used for analysis and synthesis purposes.

**Simplicity versus accuracy.** It is possible to improve the accuracy of a mathematical model by increasing its complexity. In some cases, we include hundreds of equations to describe a complete system. In obtaining a mathematical model, however, we must make a compromise between the simplicity of the model and the accuracy of the results of the analysis. If extreme accuracy is not needed, however, it is preferable to obtain only a reasonably simplified model. In fact, we are generally satisfied if we can obtain a mathematical model that is adequate for the problem under consideration. It is important to note, however, that the results obtained from the analysis are valid only to the extent that the model approximates a given dynamic system.

In deriving a reasonably simplified mathematical model, we frequently find it necessary to ignore certain inherent physical properties of the system. In particular, if a linear lumped-parameter mathematical model (that is, one employing ordinary differential equations) is desired, it is always necessary to ignore certain nonlinearities and distributed parameters (that is, ones giving rise to partial differential equations) that may be present in the physical system. If the effects that these ignored properties have on the response are small, good agreement will be obtained between the results of the analysis of a mathematical model and the results of the experimental study of the physical system.

In general, in solving a new problem, we find it desirable first to build a simplified model so that we can get a general feeling for the solution. A more complete mathematical model may then be built and used for a more complete analysis.

We must be well aware of the fact that a linear lumped-parameter model, which may be valid in low-frequency operations, may not be valid at sufficiently high frequencies since the neglected property of distributed parameters may become an important factor in the dynamic behavior of the system. For example, the mass of a spring may be neglected in low-frequency operations, but it becomes an important property of the system at high frequencies.

**Linear systems.** A system is called linear if the principle of superposition applies. The principle of superposition states that the response produced by the simultaneous application of two different forcing functions is the sum of the two individual responses. Hence, for the linear system, the response to several inputs can be calculated by treating one input at a time and adding the results. It is this principle that allows one to build up complicated solutions to the linear differential equation from simple solutions.

In an experimental investigation of a dynamic system, if cause and effect are proportional, thus implying that the principle of superposition holds, then the system can be considered linear.

**Linear time-invariant systems and linear time-varying systems.** A differential equation is linear if the coefficients are constants or functions only of the independent variable. Dynamic systems that are composed of linear time-invariant lumped-parameter components may be described by linear time-invariant (constant-coefficient) differential equations. Such systems are called *linear time-invariant* (or *linear constant-coefficient*) systems. Systems that are represented by differential equations whose coefficients are functions of time are called *linear time-varying* systems. An example of a time-varying control system is a spacecraft control system. (The mass of a spacecraft changes due to fuel consumption.)

**Nonlinear systems.** A system is nonlinear if the principle of superposition does not apply. Thus, for a nonlinear system the response to two inputs cannot be calculated by treating one input at a time and adding the results. Examples of nonlinear differential equations are

$$\frac{d^2x}{dt^2} + \left(\frac{dx}{dt}\right)^2 + x = A \sin \omega t$$

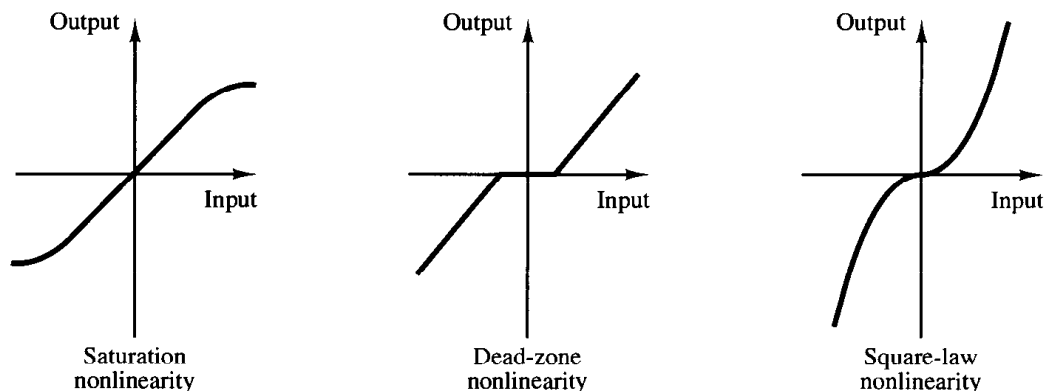
$$\frac{d^2x}{dt^2} + (x^2 - 1) \frac{dx}{dt} + x = 0$$

$$\frac{d^2x}{dt^2} + \frac{dx}{dt} + x + x^3 = 0$$

Although many physical relationships are often represented by linear equations, in most cases actual relationships are not quite linear. In fact, a careful study of physical systems reveals that even so-called “linear systems” are really linear only in limited operating ranges. In practice, many electromechanical systems, hydraulic systems, pneumatic systems, and so on, involve nonlinear relationships among the variables. For example, the output of a component may saturate for large input signals. There may be a dead space that affects small signals. (The dead space of a component is a small range of input variations to which the component is insensitive.) Square-law nonlinearity may occur in some components. For instance, dampers used in physical systems may be linear for low-velocity operations but may become nonlinear at high velocities, and the damping force may become proportional to the square of the operating velocity. Examples of characteristic curves for these nonlinearities are shown in Figure 3–1.

Note that some important control systems are nonlinear for signals of any size. For example, in on–off control systems, the control action is either on or off, and there is no linear relationship between the input and output of the controller.

Procedures for finding the solutions of problems involving such nonlinear systems, in general, are extremely complicated. Because of this mathematical difficulty attached to nonlinear systems, one often finds it necessary to introduce “equivalent” linear systems in place of nonlinear ones. Such equivalent linear systems are valid for only a limited range of operation. Once a nonlinear system is approximated by a linear mathematical model, a number of linear tools may be applied for analysis and design purposes.



**Figure 3–1**  
Characteristic curves  
for various  
nonlinearities.

**Linearization of nonlinear systems.** In control engineering a normal operation of the system may be around an equilibrium point, and the signals may be considered small signals around the equilibrium. (It should be pointed out that there are many exceptions to such a case.) However, if the system operates around an equilibrium point and if the signals involved are small signals, then it is possible to approximate the nonlinear system by a linear system. Such a linear system is equivalent to the nonlinear system considered within a limited operating range. Such a linearized model (linear, time-invariant model) is very important in control engineering. We shall discuss a linearization technique in Section 3–10.

**Outline of the chapter.** Section 3–1 has presented an introduction to the mathematical modeling of dynamic systems, including discussions of linear and nonlinear systems. Section 3–2 presents the transfer function and impulse-response function. Section 3–3 introduces block diagrams and Section 3–4 discusses concepts of modeling in state space. Section 3–5 presents state-space representation of dynamic systems. Section 3–6 treats mathematical modeling of mechanical systems. We discuss Newton's approach to modeling mechanical systems. Section 3–7 deals with mathematical modeling of electrical circuits, Section 3–8 treats liquid-level systems, and Section 3–9 presents mathematical modeling of thermal systems. Finally, Section 3–10 discusses the linearization of nonlinear mathematical models. (Mathematical modeling of other types of systems is treated throughout the remaining chapters of the book.)

### 3–2 TRANSFER FUNCTION AND IMPULSE-RESPONSE FUNCTION

In control theory, functions called transfer functions are commonly used to characterize the input–output relationships of components or systems that can be described by linear, time-invariant, differential equations. We begin by defining the transfer function and follow with a derivation of the transfer function of a mechanical system. Then we discuss the impulse-response function.

**Transfer function.** The *transfer function* of a linear, time-invariant, differential equation system is defined as the ratio of the Laplace transform of the output (response function) to the Laplace transform of the input (driving function) under the assumption that all initial conditions are zero.

Consider the linear time-invariant system defined by the following differential equation:

$$\begin{aligned} a_0 y^{(n)} + a_1 y^{(n-1)} + \cdots + a_{n-1} \dot{y} + a_n y \\ = b_0 x^{(m)} + b_1 x^{(m-1)} + \cdots + b_{m-1} \dot{x} + b_m x \quad (n \geq m) \end{aligned} \quad (3-1)$$

where  $y$  is the output of the system and  $x$  is the input. The transfer function of this system is obtained by taking the Laplace transforms of both sides of Equation (3–1), under the assumption that all initial conditions are zero, or

$$\begin{aligned}\text{Transfer function} = G(s) &= \frac{\mathcal{L}[\text{output}]}{\mathcal{L}[\text{input}]} \Big|_{\text{zero initial conditions}} \\ &= \frac{Y(s)}{X(s)} = \frac{b_0 s^m + b_1 s^{m-1} + \cdots + b_{m-1} s + b_m}{a_0 s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n}\end{aligned}\quad (3-2)$$

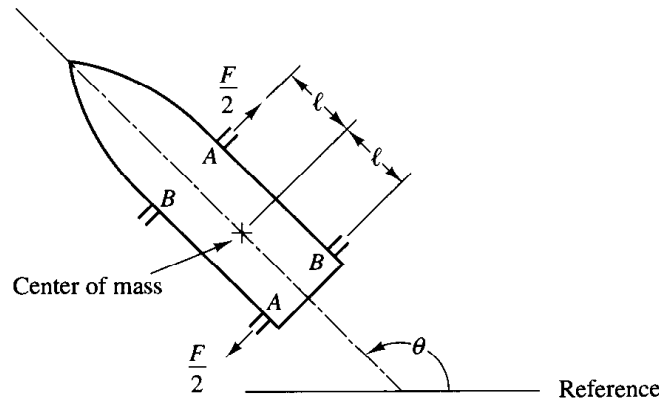
By using the concept of transfer function, it is possible to represent system dynamics by algebraic equations in  $s$ . If the highest power of  $s$  in the denominator of the transfer function is equal to  $n$ , the system is called an  *$n$ th-order system*.

**Comments on transfer function.** The applicability of the concept of the transfer function is limited to linear, time-invariant, differential equation systems. The transfer function approach, however, is extensively used in the analysis and design of such systems. In what follows, we shall list important comments concerning the transfer function. (Note that in the list a system referred to is one described by a linear, time-invariant, differential equation.)

1. The transfer function of a system is a mathematical model in that it is an operational method of expressing the differential equation that relates the output variable to the input variable.
2. The transfer function is a property of a system itself, independent of the magnitude and nature of the input or driving function.
3. The transfer function includes the units necessary to relate the input to the output; however, it does not provide any information concerning the physical structure of the system. (The transfer functions of many physically different systems can be identical.)
4. If the transfer function of a system is known, the output or response can be studied for various forms of inputs with a view toward understanding the nature of the system.
5. If the transfer function of a system is unknown, it may be established experimentally by introducing known inputs and studying the output of the system. Once established, a transfer function gives a full description of the dynamic characteristics of the system, as distinct from its physical description.

**Mechanical system.** Consider the satellite attitude control system shown in Figure 3-2. The diagram shows the control of only the yaw angle  $\theta$ . (In the actual system there are controls about three axes.) Small jets apply reaction forces to rotate the satellite body into the desired attitude. The two skew symmetrically placed jets denoted by  $A$  or  $B$  operate in pairs. Assume that each jet thrust is  $F/2$  and a torque  $T = Fl$  is applied to the system. The jets are applied for a certain time duration and thus the torque can be written as  $T(t)$ . The moment of inertia about the axis of rotation at the center of mass is  $J$ .

Let us obtain the transfer function of this system by assuming that torque  $T(t)$  is the input, and the angular displacement  $\theta(t)$  of the satellite is the output. (We consider the motion only in the plane of the page.)



**Figure 3-2**  
Schematic diagram of a satellite attitude control system.

To derive the transfer function, we proceed according to the following steps.

1. Write the differential equation for the system.
2. Take the Laplace transform of the differential equation, assuming all initial conditions are zero.
3. Take the ratio of the output  $\Theta(s)$  to the input  $T(s)$ . This ratio is the transfer function.

Applying Newton's second law to the present system and noting that there is no friction in the environment of the satellite, we have

$$J \frac{d^2\theta}{dt^2} = T$$

Taking the Laplace transform of both sides of this last equation and assuming all initial conditions to be zero yields

$$Js^2\Theta(s) = T(s)$$

where  $\Theta(s) = \mathcal{L}[\theta(t)]$  and  $T(s) = \mathcal{L}[T(t)]$ . The transfer function of the system is thus obtained as

$$\text{Transfer function} = \frac{\Theta(s)}{T(s)} = \frac{1}{Js^2}$$

**Convolution integral.** For a linear, time-invariant system the transfer function  $G(s)$  is

$$G(s) = \frac{Y(s)}{X(s)}$$

where  $X(s)$  is the Laplace transform of the input and  $Y(s)$  is the Laplace transform of the output, where we assume that all initial conditions involved are zero. It follows that the output  $Y(s)$  can be written as the product of  $G(s)$  and  $X(s)$ , or

$$Y(s) = G(s)X(s) \quad (3-3)$$

Note that multiplication in the complex domain is equivalent to convolution in the time domain, so the inverse Laplace transform of Equation (3-3) is given by the following convolution integral:

$$\begin{aligned}
 y(t) &= \int_0^t x(\tau)g(t - \tau) d\tau \\
 &= \int_0^t g(\tau)x(t - \tau) d\tau
 \end{aligned} \tag{3-4}$$

where  $g(t) = 0$  and  $x(t) = 0$  for  $t < 0$ .

**Impulse-response function.** Consider the output (response) of a system to a unit-impulse input when the initial conditions are zero. Since the Laplace transform of the unit-impulse function is unity, the Laplace transform of the output of the system is

$$Y(s) = G(s) \tag{3-5}$$

The inverse Laplace transform of the output given by Equation (3-5) gives the impulse response of the system. The inverse Laplace transform of  $G(s)$ , or

$$\mathcal{L}^{-1}[G(s)] = g(t)$$

is called the impulse-response function. This function  $g(t)$  is also called the weighting function of the system.

The impulse-response function  $g(t)$  is thus the response of a linear system to a unit-impulse input when the initial conditions are zero. The Laplace transform of this function gives the transfer function. Therefore, the transfer function and impulse-response function of a linear, time-invariant system contain the same information about the system dynamics. It is hence possible to obtain complete information about the dynamic characteristics of the system by exciting it with an impulse input and measuring the response. (In practice, a pulse input with a very short duration compared with the significant time constants of the system can be considered an impulse.)

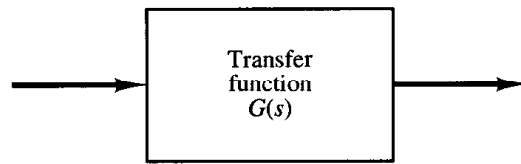
### 3-3 BLOCK DIAGRAMS

A control system may consist of a number of components. To show the functions performed by each component, in control engineering, we commonly use a diagram called the *block diagram*. This section explains what a block diagram is, presents a method for obtaining block diagrams for physical systems, and, finally, discusses techniques to simplify such diagrams.

**Block diagrams.** A *block diagram* of a system is a pictorial representation of the functions performed by each component and of the flow of signals. Such a diagram depicts the interrelationships that exist among the various components. Differing from a purely abstract mathematical representation, a block diagram has the advantage of indicating more realistically the signal flows of the actual system.

In a block diagram all system variables are linked to each other through functional blocks. The *functional block* or simply *block* is a symbol for the mathematical operation on the input signal to the block that produces the output. The transfer functions of the components are usually entered in the corresponding blocks, which are connected by arrows to indicate the direction of the flow of signals. Note that the signal can pass only





**Figure 3-3**  
Element of a block diagram.

in the direction of the arrows. Thus a block diagram of a control system explicitly shows a unilateral property.

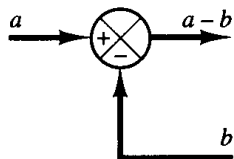
Figure 3-3 shows an element of the block diagram. The arrowhead pointing toward the block indicates the input, and the arrowhead leading away from the block represents the output. Such arrows are referred to as *signals*.

Note that the dimensions of the output signal from the block are the dimensions of the input signal multiplied by the dimensions of the transfer function in the block.

The advantages of the block diagram representation of a system lie in the fact that it is easy to form the overall block diagram for the entire system by merely connecting the blocks of the components according to the signal flow and that it is possible to evaluate the contribution of each component to the overall performance of the system.

In general, the functional operation of the system can be visualized more readily by examining the block diagram than by examining the physical system itself. A block diagram contains information concerning dynamic behavior, but it does not include any information on the physical construction of the system. Consequently, many dissimilar and unrelated systems can be represented by the same block diagram.

It should be noted that in a block diagram the main source of energy is not explicitly shown and that the block diagram of a given system is not unique. A number of different block diagrams can be drawn for a system, depending on the point of view of the analysis.



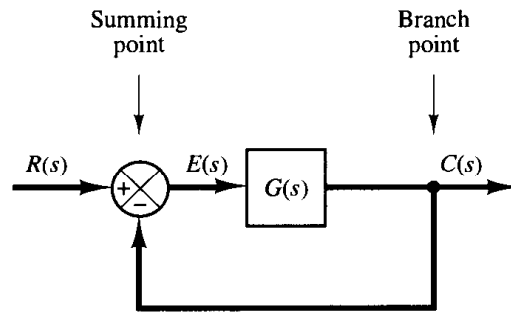
**Figure 3-4**  
Summing point.

**Summing Point.** Referring to Figure 3-4, a circle with a cross is the symbol that indicates a summing operation. The plus or minus sign at each arrowhead indicates whether that signal is to be added or subtracted. It is important that the quantities being added or subtracted have the same dimensions and the same units.

**Branch Point.** A *branch point* is a point from which the signal from a block goes concurrently to other blocks or summing points.

**Block diagram of a closed-loop system.** Figure 3-5 shows an example of a block diagram of a closed-loop system. The output  $C(s)$  is fed back to the summing point, where it is compared with the reference input  $R(s)$ . The closed-loop nature of the system is clearly indicated by the figure. The output of the block,  $C(s)$  in this case, is obtained by multiplying the transfer function  $G(s)$  by the input to the block,  $E(s)$ . Any linear control system may be represented by a block diagram consisting of blocks, summing points, and branch points.

When the output is fed back to the summing point for comparison with the input, it is necessary to convert the form of the output signal to that of the input signal. For example, in a temperature-control system, the output signal is usually the controlled temperature. The output signal, which has the dimension of temperature, must be converted



**Figure 3–5**  
Block diagram of a closed-loop system.

to a force or position or voltage before it can be compared with the input signal. This conversion is accomplished by the feedback element whose transfer function is  $H(s)$ , as shown in Figure 3–6. The role of the feedback element is to modify the output before it is compared with the input. (In most cases the feedback element is a sensor that measures the output of the plant. The output of the sensor is compared with the input, and the actuating error signal is generated.) In the present example, the feedback signal that is fed back to the summing point for comparison with the input is  $B(s) = H(s)C(s)$ .

**Open-loop transfer function and feedforward transfer function.** Referring to Figure 3–6, the ratio of the feedback signal  $B(s)$  to the actuating error signal  $E(s)$  is called the *open-loop transfer function*. That is,

$$\text{Open-loop transfer function} = \frac{B(s)}{E(s)} = G(s)H(s)$$

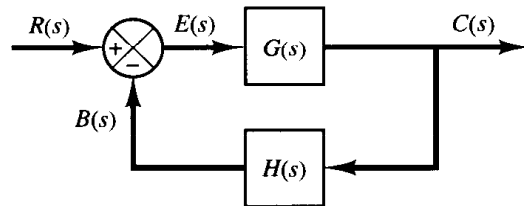
The ratio of the output  $C(s)$  to the actuating error signal  $E(s)$  is called the *feedforward transfer function*, so that

$$\text{Feedforward transfer function} = \frac{C(s)}{E(s)} = G(s)$$

If the feedback transfer function  $H(s)$  is unity, then the open-loop transfer function and the feedforward transfer function are the same.

**Closed-loop transfer function.** For the system shown in Figure 3–6, the output  $C(s)$  and input  $R(s)$  are related as follows:

$$\begin{aligned} C(s) &= G(s)E(s) \\ E(s) &= R(s) - B(s) \\ &= R(s) - H(s)C(s) \end{aligned}$$



**Figure 3–6**  
Closed-loop system.

Eliminating  $E(s)$  from these equations gives

$$C(s) = G(s)[R(s) - H(s)C(s)]$$

or

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)} \quad (3-6)$$

The transfer function relating  $C(s)$  to  $R(s)$  is called the *closed-loop transfer function*. This transfer function relates the closed-loop system dynamics to the dynamics of the feedforward elements and feedback elements.

From Equation (3-6),  $C(s)$  is given by

$$C(s) = \frac{G(s)}{1 + G(s)H(s)} R(s)$$

Thus the output of the closed-loop system clearly depends on both the closed-loop transfer function and the nature of the input.

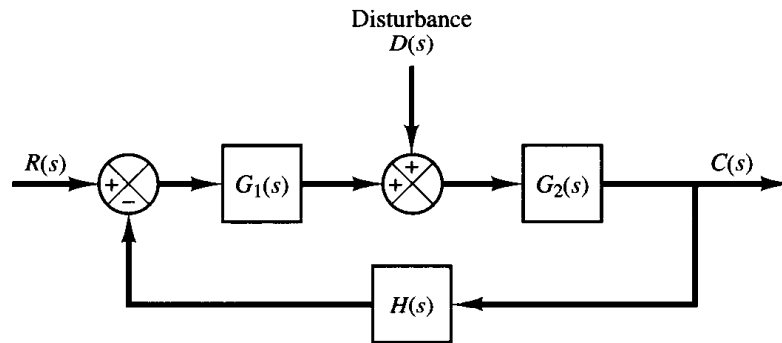
**Closed-loop system subjected to a disturbance.** Figure 3-7 shows a closed-loop system subjected to a disturbance. When two inputs (the reference input and disturbance) are present in a linear system, each input can be treated independently of the other; and the outputs corresponding to each input alone can be added to give the complete output. The way each input is introduced into the system is shown at the summing point by either a plus or minus sign.

Consider the system shown in Figure 3-7. In examining the effect of the disturbance  $D(s)$ , we may assume that the system is at rest initially with zero error; we may then calculate the response  $C_D(s)$  to the disturbance only. This response can be found from

$$\frac{C_D(s)}{D(s)} = \frac{G_2(s)}{1 + G_1(s)G_2(s)H(s)}$$

On the other hand, in considering the response to the reference input  $R(s)$ , we may assume that the disturbance is zero. Then the response  $C_R(s)$  to the reference input  $R(s)$  can be obtained from

$$\frac{C_R(s)}{R(s)} = \frac{G_1(s)G_2(s)}{1 + G_1(s)G_2(s)H(s)}$$



**Figure 3-7**  
Closed-loop system  
subjected to a  
disturbance.

The response to the simultaneous application of the reference input and disturbance can be obtained by adding the two individual responses. In other words, the response  $C(s)$  due to the simultaneous application of the reference input  $R(s)$  and disturbance  $D(s)$  is given by

$$\begin{aligned} C(s) &= C_R(s) + C_D(s) \\ &= \frac{G_2(s)}{1 + G_1(s)G_2(s)H(s)} [G_1(s)R(s) + D(s)] \end{aligned}$$

Consider now the case where  $|G_1(s)H(s)| \gg 1$  and  $|G_1(s)G_2(s)H(s)| \gg 1$ . In this case, the closed-loop transfer function  $C_D(s)/D(s)$  becomes almost zero, and the effect of the disturbance is suppressed. This is an advantage of the closed-loop system.

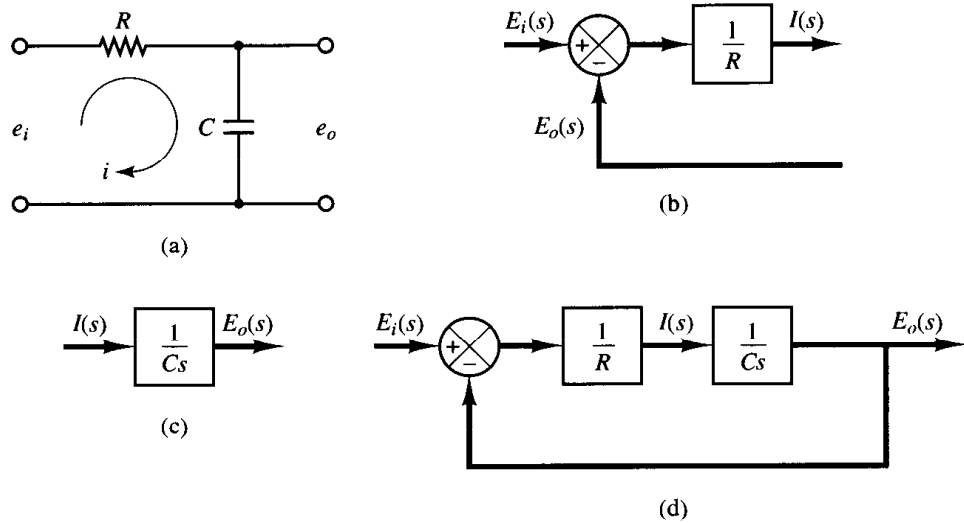
On the other hand, the closed-loop transfer function  $C_R(s)/R(s)$  approaches  $1/H(s)$  as the gain of  $G_1(s)G_2(s)H(s)$  increases. This means that if  $|G_1(s)G_2(s)H(s)| \gg 1$  then the closed-loop transfer function  $C_R(s)/R(s)$  becomes independent of  $G_1(s)$  and  $G_2(s)$  and becomes inversely proportional to  $H(s)$  so that the variations of  $G_1(s)$  and  $G_2(s)$  do not affect the closed-loop transfer function  $C_R(s)/R(s)$ . This is another advantage of the closed-loop system. It can easily be seen that any closed-loop system with unity feedback,  $H(s) = 1$ , tends to equalize the input and output.

**Procedures for drawing a block diagram.** To draw a block diagram for a system, first write the equations that describe the dynamic behavior of each component. Then take the Laplace transforms of these equations, assuming zero initial conditions, and represent each Laplace-transformed equation individually in block form. Finally, assemble the elements into a complete block diagram.

As an example, consider the  $RC$  circuit shown in Figure 3–8(a). The equations for this circuit are

$$i = \frac{e_i - e_o}{R} \quad (3-7)$$

$$e_o = \frac{\int i dt}{C} \quad (3-8)$$



**Figure 3–8**  
(a)  $RC$  circuit; (b) block diagram representing Equation (3–9); (c) block diagram representing Equation (3–10); (d) block diagram of the  $RC$  circuit.

The Laplace transforms of Equations (3-7) and (3-8), with zero initial condition, become

$$I(s) = \frac{E_i(s) - E_o(s)}{R} \quad (3-9)$$

$$E_o(s) = \frac{I(s)}{Cs} \quad (3-10)$$

Equation (3-9) represents a summing operation, and the corresponding diagram is shown in Figure 3-8(b). Equation (3-10) represents the block as shown in Figure 3-8(c). Assembling these two elements, we obtain the overall block diagram for the system as shown in Figure 3-8(d).

**Block diagram reduction.** It is important to note that blocks can be connected in series only if the output of one block is not affected by the next following block. If there are any loading effects between the components, it is necessary to combine these components into a single block.

Any number of cascaded blocks representing nonloading components can be replaced by a single block, the transfer function of which is simply the product of the individual transfer functions.

A complicated block diagram involving many feedback loops can be simplified by a step-by-step rearrangement, using rules of block diagram algebra. Some of these important rules are given in Table 3-1. They are obtained by writing the same equation in

**Table 3-1** Rules of Block Diagram Algebra

|   | Original Block Diagrams | Equivalent Block Diagrams |
|---|-------------------------|---------------------------|
| 1 |                         |                           |
| 2 |                         |                           |
| 3 |                         |                           |
| 4 |                         |                           |
| 5 |                         |                           |

a different way. Simplification of the block diagram by rearrangements and substitutions considerably reduces the labor needed for subsequent mathematical analysis. It should be noted, however, that as the block diagram is simplified the transfer functions in new blocks become more complex because new poles and new zeros are generated.

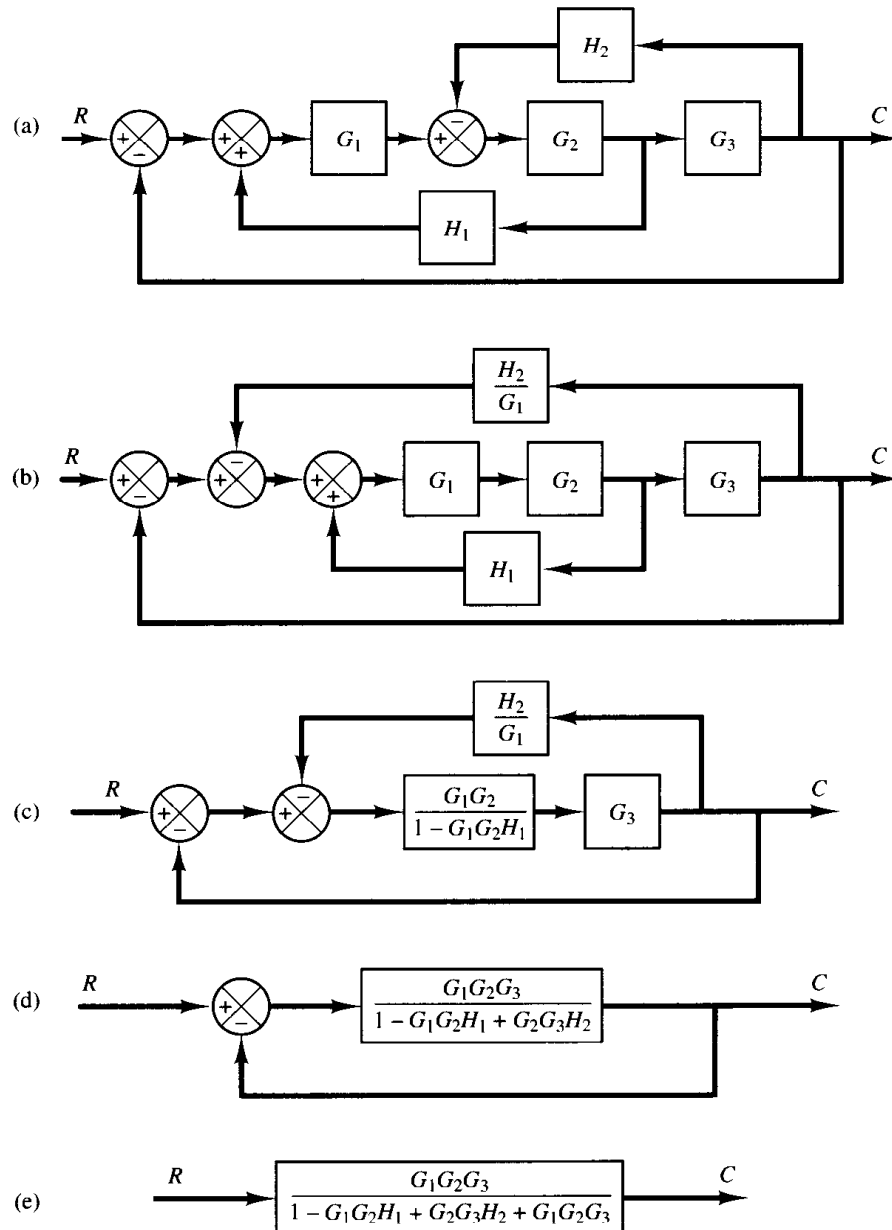
In simplifying a block diagram, remember the following.

1. The product of the transfer functions in the feedforward direction must remain the same.
2. The product of the transfer functions around the loop must remain the same.

### EXAMPLE 3-1

Consider the system shown in Figure 3-9(a). Simplify this diagram.

By moving the summing point of the negative feedback loop containing  $H_2$  outside the positive feedback loop containing  $H_1$ , we obtain Figure 3-9(b). Eliminating the positive feedback



**Figure 3-9**  
(a) Multiple-loop system; (b)–(e) successive reductions of the block diagram shown in (a).

loop, we have Figure 3–9(c). The elimination of the loop containing  $H_2/G_1$  gives Figure 3–9(d). Finally, eliminating the feedback loop results in Figure 3–9(e).

Notice that the numerator of the closed-loop transfer function  $C(s)/R(s)$  is the product of the transfer functions of the feedforward path. The denominator of  $C(s)/R(s)$  is equal to

$$\begin{aligned} 1 - \sum (\text{product of the transfer functions around each loop}) \\ &= 1 - (G_1 G_2 H_1 - G_2 G_3 H_2 - G_1 G_2 G_3) \\ &= 1 - G_1 G_2 H_1 + G_2 G_3 H_2 + G_1 G_2 G_3 \end{aligned}$$

(The positive feedback loop yields a negative term in the denominator.)

### 3–4 MODELING IN STATE SPACE

In this section we shall present introductory material on state-space analysis of control systems.

**Modern control theory.** The modern trend in engineering systems is toward greater complexity, due mainly to the requirements of complex tasks and good accuracy. Complex systems may have multiple inputs and multiple outputs and may be time varying. Because of the necessity of meeting increasingly stringent requirements on the performance of control systems, the increase in system complexity, and easy access to large-scale computers, modern control theory, which is a new approach to the analysis and design of complex control systems, has been developed since around 1960. This new approach is based on the concept of state. The concept of state by itself is not new since it has been in existence for a long time in the field of classical dynamics and other fields.

**Modern control theory versus conventional control theory.** Modern control theory is contrasted with conventional control theory in that the former is applicable to multiple-input–multiple-output systems, which may be linear or nonlinear, time invariant or time varying, while the latter is applicable only to linear time-invariant single-input–single-output systems. Also, modern control theory is essentially a time-domain approach, while conventional control theory is a complex frequency-domain approach. Before we proceed further, we must define state, state variables, state vector, and state space.

**State.** The state of a dynamic system is the smallest set of variables (called *state variables*) such that the knowledge of these variables at  $t = t_0$ , together with the knowledge of the input for  $t \geq t_0$ , completely determines the behavior of the system for any time  $t \geq t_0$ .

Note that the concept of state is by no means limited to physical systems. It is applicable to biological systems, economic systems, social systems, and others.

**State variables.** The state variables of a dynamic system are the variables making up the smallest set of variables that determine the state of the dynamic system. If at least  $n$  variables  $x_1, x_2, \dots, x_n$  are needed to completely describe the behavior of a dy-

dynamic system (so that once the input is given for  $t \geq t_0$  and the initial state at  $t = t_0$  is specified, the future state of the system is completely determined), then such  $n$  variables are a set of state variables.

Note that state variables need not be physically measurable or observable quantities. Variables that do not represent physical quantities and those that are neither measurable nor observable can be chosen as state variables. Such freedom in choosing state variables is an advantage of the state-space methods. Practically, however, it is convenient to choose easily measurable quantities for the state variables, if this is possible at all, because optimal control laws will require the feedback of all state variables with suitable weighting.

**State vector.** If  $n$  state variables are needed to completely describe the behavior of a given system, then these  $n$  state variables can be considered the  $n$  components of a vector  $\mathbf{x}$ . Such a vector is called a *state vector*. A state vector is thus a vector that determines uniquely the system state  $\mathbf{x}(t)$  for any time  $t \geq t_0$ , once the state at  $t = t_0$  is given and the input  $\mathbf{u}(t)$  for  $t \geq t_0$  is specified.

**State space.** The  $n$ -dimensional space whose coordinate axes consist of the  $x_1$  axis,  $x_2$  axis,  $\dots$ ,  $x_n$  axis is called a *state space*. Any state can be represented by a point in the state space.

**State-space equations.** In state-space analysis we are concerned with three types of variables that are involved in the modeling of dynamic systems: input variables, output variables, and state variables. As we shall see in Section 3–5, the state-space representation for a given system is not unique, except that the number of state variables is the same for any of the different state-space representations of the same system.

The dynamic system must involve elements that memorize the values of the input for  $t \geq t_1$ . Since integrators in a continuous-time control system serve as memory devices, the outputs of such integrators can be considered as the variables that define the internal state of the dynamic system. Thus the outputs of integrators serve as state variables. The number of state variables to completely define the dynamics of the system is equal to the number of integrators involved in the system.

Assume that a multiple-input–multiple-output system involves  $n$  integrators. Assume also that there are  $r$  inputs  $u_1, u_2(t), \dots, u_r(t)$  and  $m$  outputs  $y_1(t), y_2(t), \dots, y_m(t)$ . Define  $n$  outputs of the integrators as state variables:  $x_1(t), x_2(t), \dots, x_n(t)$ . Then the system may be described by

$$\begin{aligned}\dot{x}_1(t) &= f_1(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_r; t) \\ \dot{x}_2(t) &= f_2(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_r; t) \\ &\vdots \\ \dot{x}_n(t) &= f_n(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_r; t)\end{aligned}\tag{3-11}$$



The outputs  $y_1(t), y_2(t), \dots, y_m(t)$  of the system may be given by

$$\begin{aligned} y_1(t) &= g_1(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_r; t) \\ y_2(t) &= g_2(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_r; t) \\ &\vdots \\ y_m(t) &= g_m(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_r; t) \end{aligned} \quad (3-12)$$

If we define

$$\begin{aligned} \mathbf{x}(t) &= \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}, \quad \mathbf{f}(\mathbf{x}, \mathbf{u}, t) = \begin{bmatrix} f_1(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_r; t) \\ f_2(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_r; t) \\ \vdots \\ f_n(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_r; t) \end{bmatrix} \\ \mathbf{y}(t) &= \begin{bmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_m(t) \end{bmatrix}, \quad \mathbf{g}(\mathbf{x}, \mathbf{u}, t) = \begin{bmatrix} g_1(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_r; t) \\ g_2(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_r; t) \\ \vdots \\ g_m(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_r; t) \end{bmatrix}, \quad \mathbf{u}(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \\ \vdots \\ u_r(t) \end{bmatrix} \end{aligned}$$

then Equations (3-11) and (3-12) become

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}, \mathbf{u}, t) \quad (3-13)$$

$$\mathbf{y}(t) = \mathbf{g}(\mathbf{x}, \mathbf{u}, t) \quad (3-14)$$

where Equation (3-13) is the state equation and Equation (3-14) is the output equation. If vector functions  $\mathbf{f}$  and/or  $\mathbf{g}$  involve time  $t$  explicitly, then the system is called a time-varying system.

If Equations (3-13) and (3-14) are linearized about the operating state, then we have the following linearized state equation and output equation:

$$\dot{\mathbf{x}}(t) = \mathbf{A}(t)\mathbf{x}(t) + \mathbf{B}(t)\mathbf{u}(t) \quad (3-15)$$

$$\mathbf{y}(t) = \mathbf{C}(t)\mathbf{x}(t) + \mathbf{D}(t)\mathbf{u}(t) \quad (3-16)$$

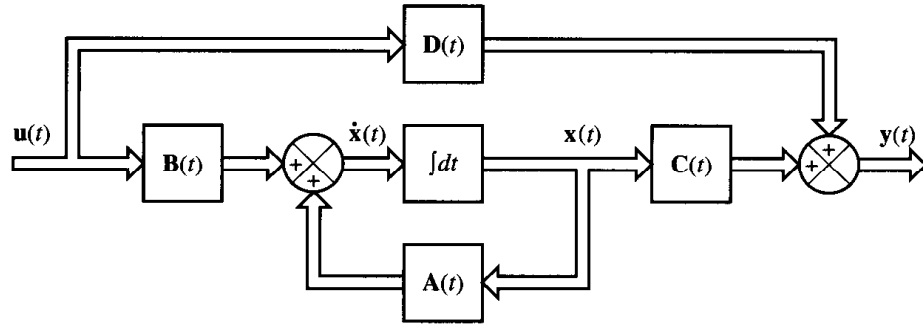
where  $\mathbf{A}(t)$  is called the state matrix,  $\mathbf{B}(t)$  the input matrix,  $\mathbf{C}(t)$  the output matrix, and  $\mathbf{D}(t)$  the direct transmission matrix. (Details of linearization of nonlinear systems about the operating state are discussed in Section 3-10.) A block diagram representation of Equations (3-15) and (3-16) is shown in Figure 3-10.

If vector functions  $\mathbf{f}$  and  $\mathbf{g}$  do not involve time  $t$  explicitly then the system is called a time-invariant system. In this case, Equations (3-15) and (3-16) can be simplified to

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \quad (3-17)$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t) \quad (3-18)$$

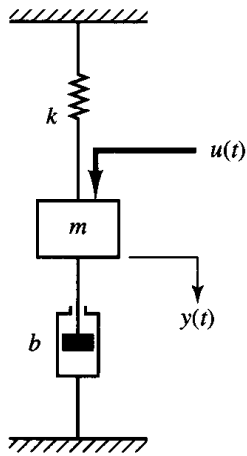
**Figure 3-10**  
Block diagram of the  
linear continuous-  
time control system  
represented in state  
space.



Equation (3-17) is the state equation of the linear, time-invariant system. Equation (3-18) is the output equation for the same system. In this book we shall be concerned mostly with systems described by Equations (3-17) and (3-18).

In what follows we shall present an example for deriving a state equation and output equation.

### EXAMPLE 3-2



**Figure 3-11**  
Mechanical system.

Consider the mechanical system shown in Figure 3-11. We assume that the system is linear. The external force  $u(t)$  is the input to the system, and the displacement  $y(t)$  of the mass is the output. The displacement  $y(t)$  is measured from the equilibrium position in the absence of the external force. This system is a single-input-single-output system.

From the diagram, the system equation is

$$m\ddot{y} + b\dot{y} + ky = u \quad (3-19)$$

This system is of second order. This means that the system involves two integrators. Let us define state variables  $x_1(t)$  and  $x_2(t)$  as

$$x_1(t) = y(t)$$

$$x_2(t) = \dot{y}(t)$$

Then we obtain

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = \frac{1}{m}(-ky - b\dot{y}) + \frac{1}{m}u$$

or

$$\dot{x}_1 = x_2 \quad (3-20)$$

$$\dot{x}_2 = -\frac{k}{m}x_1 - \frac{b}{m}x_2 + \frac{1}{m}u \quad (3-21)$$

The output equation is

$$y = x_1 \quad (3-22)$$

In a vector-matrix form, Equations (3-20) and (3-21) can be written as

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} u \quad (3-23)$$

The output equation, Equation (3-22), can be written as

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad (3-24)$$

Equation (3-23) is a state equation and Equation (3-24) is an output equation for the system. Equations (3-23) and (3-24) are in the standard form:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$$

$$y = \mathbf{C}\mathbf{x} + Du$$

where

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D = 0$$

Figure 3-12 is a block diagram for the system. Notice that the outputs of the integrators are state variables.

**Correlation between transfer functions and state-space equations.** In what follows we shall show how to derive the transfer function of a single-input-single-output system from the state-space equations.

Let us consider the system whose transfer function is given by

$$\frac{Y(s)}{U(s)} = G(s) \quad (3-25)$$

This system may be represented in state space by the following equations:

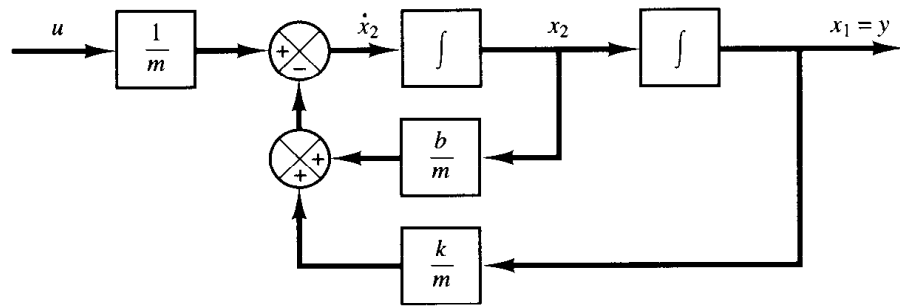
$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u \quad (3-26)$$

$$y = \mathbf{C}\mathbf{x} + Du \quad (3-27)$$

where  $\mathbf{x}$  is the state vector,  $u$  is the input, and  $y$  is the output. The Laplace transforms of Equations (3-26) and (3-27) are given by

$$s\mathbf{X}(s) - \mathbf{x}(0) = \mathbf{A}\mathbf{X}(s) + \mathbf{B}U(s) \quad (3-28)$$

$$Y(s) = \mathbf{C}\mathbf{X}(s) + DU(s) \quad (3-29)$$



**Figure 3-12**  
Block diagram of  
the mechanical  
system shown in  
Figure 3-11.

Since the transfer function was previously defined as the ratio of the Laplace transform of the output to the Laplace transform of the input when the initial conditions were zero, we assume that  $\mathbf{x}(0)$  in Equation (3-28) is zero. Then we have

$$s\mathbf{X}(s) - \mathbf{A}\mathbf{X}(s) = \mathbf{B}U(s)$$

or

$$(s\mathbf{I} - \mathbf{A})\mathbf{X}(s) = \mathbf{B}U(s)$$

By premultiplying  $(s\mathbf{I} - \mathbf{A})^{-1}$  to both sides of this last equation, we obtain

$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}U(s) \quad (3-30)$$

By substituting Equation (3-30) into Equation (3-29), we get

$$Y(s) = [\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + D]U(s) \quad (3-31)$$

Upon comparing Equation (3-31) with Equation (3-25), we see that

$$G(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + D \quad (3-32)$$

This is the transfer-function expression in terms of  $\mathbf{A}$ ,  $\mathbf{B}$ ,  $\mathbf{C}$ , and  $D$ .

Note that the right-hand side of equation (3-32) involves  $(s\mathbf{I} - \mathbf{A})^{-1}$ . Hence  $G(s)$  can be written as

$$G(s) = \frac{Q(s)}{|s\mathbf{I} - \mathbf{A}|}$$

where  $Q(s)$  is a polynomial in  $s$ . Therefore,  $|s\mathbf{I} - \mathbf{A}|$  is equal to the characteristic polynomial of  $G(s)$ . In other words, the eigenvalues of  $\mathbf{A}$  are identical to the poles of  $G(s)$ .

### EXAMPLE 3-3

Consider again the mechanical system shown in Figure 3-11. State-space equations for the system are given by Equations (3-23) and (3-24). We shall obtain the transfer function for the system from the state-space equations.

By substituting  $\mathbf{A}$ ,  $\mathbf{B}$ ,  $\mathbf{C}$ , and  $D$  into Equation (3-32), we obtain

$$\begin{aligned} G(s) &= \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + D \\ &= [1 \quad 0] \left\{ \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix} \right\}^{-1} \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} + 0 \\ &= [1 \quad 0] \begin{bmatrix} s & -1 \\ \frac{k}{m} & s + \frac{b}{m} \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} \end{aligned}$$

Since

$$\begin{bmatrix} s & -1 \\ \frac{k}{m} & s + \frac{b}{m} \end{bmatrix}^{-1} = \frac{1}{s^2 + \frac{b}{m}s + \frac{k}{m}} \begin{bmatrix} s + \frac{b}{m} & 1 \\ -\frac{k}{m} & s \end{bmatrix}$$

we have

$$G(s) = [1 \quad 0] \frac{1}{s^2 + \frac{b}{m}s + \frac{k}{m}} \begin{bmatrix} s + \frac{b}{m} & 1 \\ -\frac{k}{m} & s \end{bmatrix} \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix}$$

$$= \frac{1}{ms^2 + bs + k}$$

which is the transfer function of the system. The same transfer function can be obtained from Equation (3-19).

**Transfer matrix.** Next, consider a multiple-input-multiple-output system. Assume that there are  $r$  inputs  $u_1, u_2, \dots, u_r$  and  $m$  outputs  $y_1, y_2, \dots, y_m$ . Define

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_r \end{bmatrix}$$

The transfer matrix  $\mathbf{G}(s)$  relates the output  $\mathbf{Y}(s)$  to the input  $\mathbf{U}(s)$ , or

$$\mathbf{Y}(s) = \mathbf{G}(s)\mathbf{U}(s) \quad (3-33)$$

Since the input vector  $\mathbf{u}$  is  $r$  dimensional and the output vector  $\mathbf{y}$  is  $m$  dimensional, the transfer matrix is an  $m \times r$  matrix.

### 3-5 STATE-SPACE REPRESENTATION OF DYNAMIC SYSTEMS

A dynamic system consisting of a finite number of lumped elements may be described by ordinary differential equations in which time is the independent variable. By use of vector-matrix notation, an  $n$ th-order differential equation may be expressed by a first-order vector-matrix differential equation. If  $n$  elements of the vector are a set of state variables, then the vector-matrix differential equation is a *state* equation. In this section we shall present methods for obtaining state-space representations of continuous-time systems.

**State-space representation of  $n$ th-order systems of linear differential equations in which the forcing function does not involve derivative terms.** Consider the following  $n$ th-order system:

$$y^{(n)} + a_1 y^{(n-1)} + \dots + a_{n-1} \dot{y} + a_n y = u \quad (3-34)$$

Noting that the knowledge of  $y(0), \dot{y}(0), \dots, y^{(n-1)}(0)$ , together with the input  $u(t)$  for  $t \geq 0$ , determines completely the future behavior of the system, we may take  $y(t), \dot{y}(t), \dots, y^{(n-1)}(t)$  as a set of  $n$  state variables. (Mathematically, such a choice of state vari-

ables is quite convenient. Practically, however, because higher-order derivative terms are inaccurate, due to the noise effects inherent in any practical situations, such a choice of the state variables may not be desirable.)

Let us define

$$\begin{aligned}x_1 &= y \\x_2 &= \dot{y} \\&\vdots \\&\vdots \\x_n &= y^{(n-1)}\end{aligned}$$

Then Equation (3-34) can be written as

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_3 \\ &\vdots \\ \dot{x}_{n-1} &= x_n \\ \dot{x}_n &= -a_n x_1 - \cdots - a_1 x_n + u\end{aligned}$$

or

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u \quad (3-35)$$

where

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ \vdots & \vdots & \vdots & & \vdots \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

The output can be given by

$$y = [1 \quad 0 \quad \cdots \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ \vdots \\ x_n \end{bmatrix}$$

or

$$y = \mathbf{C}\mathbf{x} \quad (3-36)$$

where

$$\mathbf{C} = [1 \quad 0 \quad \cdots \quad 0]$$

[Note that  $D$  in Equation (3-27) is zero.] The first-order differential equation, Equation (3-35), is the state equation, and the algebraic equation, Equation (3-36), is the output equation. A block diagram realization of the state equation and output equation given by Equations (3-35) and (3-36), respectively, is shown in Figure 3-13.

Note that the state-space representation for the transfer function system

$$\frac{Y(s)}{U(s)} = \frac{1}{s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n}$$

is given also by Equations (3-35) and (3-36).

**State-space representation of  $n$ th-order systems of linear differential equations in which the forcing function involves derivative terms.** If the differential equation of the system involves derivatives of the forcing function, such as

$$y^{(n)} + a_1 y^{(n-1)} + \cdots + a_{n-1} \dot{y} + a_n y = b_0 u^{(n)} + b_1 u^{(n-1)} + \cdots + b_{n-1} \dot{u} + b_n u \quad (3-37)$$

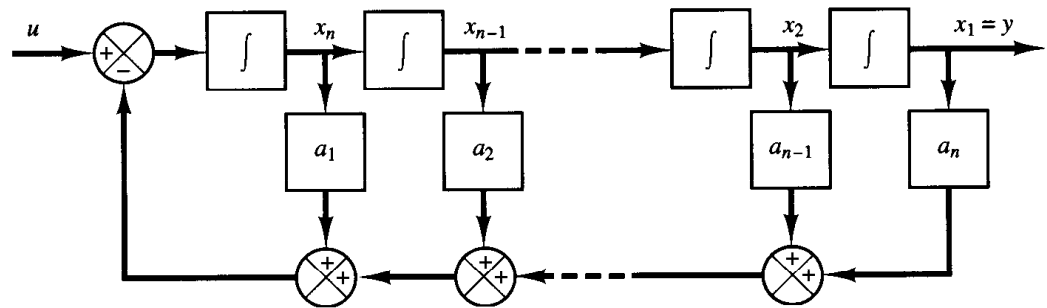
then the set of  $n$  variables  $y, \dot{y}, \ddot{y}, \dots, y^{(n-1)}$  does not qualify as a set of state variables, and the straightforward method previously employed cannot be used. This is because  $n$  first-order differential equations

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_3 \\ &\vdots \\ \dot{x}_n &= -a_n x_1 - a_{n-1} x_2 - \cdots - a_1 x_n + b_0 u^{(n)} + b_1 u^{(n-1)} + \cdots + b_n u \end{aligned}$$

where  $x_1 = y$ , may not yield a unique solution.

The main problem in defining the state variables for this case lies in the derivative terms on the right-hand side of the last of the preceding  $n$  equations. The state variables must be such that they will eliminate the derivatives of  $u$  in the state equation.

One way to obtain a state equation and output equation is to define the following  $n$  variables as a set of  $n$  state variables:



**Figure 3-13**  
Block diagram realization of state equation and output equation given by Equations (3-35) and (3-36), respectively.

$$\begin{aligned}
x_1 &= y - \beta_0 u \\
x_2 &= \dot{y} - \beta_0 \dot{u} - \beta_1 u = \dot{x}_1 - \beta_1 u \\
x_3 &= \ddot{y} - \beta_0 \ddot{u} - \beta_1 \dot{u} - \beta_2 u = \dot{x}_2 - \beta_2 u \\
&\vdots \\
x_n &= y^{(n-1)} - \beta_0^{(n-1)} u - \beta_1^{(n-1)} \dot{u} - \cdots - \beta_{n-2} \dot{u} - \beta_{n-1} u = \dot{x}_{n-1} - \beta_{n-1} u
\end{aligned} \tag{3-38}$$

where  $\beta_0, \beta_1, \beta_2, \dots, \beta_n$  are determined from

$$\begin{aligned}
\beta_0 &= b_0 \\
\beta_1 &= b_1 - a_1 \beta_0 \\
\beta_2 &= b_2 - a_1 \beta_1 - a_2 \beta_0 \\
\beta_3 &= b_3 - a_1 \beta_2 - a_2 \beta_1 - a_3 \beta_0 \\
&\vdots \\
\beta_n &= b_n - a_1 \beta_{n-1} - \cdots - a_{n-1} \beta_1 - a_n \beta_0
\end{aligned} \tag{3-39}$$

With this choice of state variables the existence and uniqueness of the solution of the state equation is guaranteed. (Note that this is not the only choice of a set of state variables.) With the present choice of state variables, we obtain

$$\begin{aligned}
\dot{x}_1 &= x_2 + \beta_1 u \\
\dot{x}_2 &= x_3 + \beta_2 u \\
&\vdots \\
\dot{x}_{n-1} &= x_n + \beta_{n-1} u \\
\dot{x}_n &= -a_n x_1 - a_{n-1} x_2 - \cdots - a_1 x_n + \beta_n u
\end{aligned} \tag{3-40}$$

[To derive Equation (3-40), see Problem A-3-3.] In terms of vector-matrix equations, Equation (3-40) and the output equation can be written as

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_{n-1} \\ \beta_n \end{bmatrix} u$$

$$y = [1 \quad 0 \quad \cdots \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \beta_0 u$$



or

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u \quad (3-41)$$

$$y = \mathbf{C}\mathbf{x} + Du \quad (3-42)$$

where

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix}$$

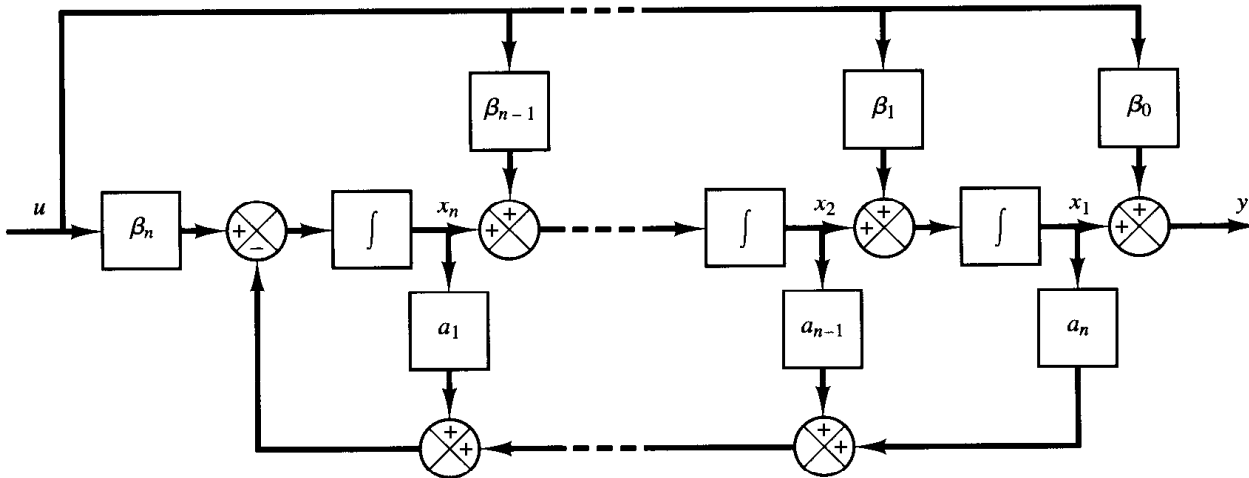
$$\mathbf{B} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_{n-1} \\ \beta_n \end{bmatrix}, \quad \mathbf{C} = [1 \ 0 \ \cdots \ 0], \quad D = \beta_0 = b_0$$

The initial condition  $\mathbf{x}(0)$  may be determined by use of Equation (3-38).

In this state-space representation, matrices  $\mathbf{A}$  and  $\mathbf{C}$  are exactly the same as those for the system of Equation (3-34). The derivatives on the right-hand side of Equation (3-37) affect only the elements of the  $\mathbf{B}$  matrix.

Note that the state-space representation for the transfer function

$$\frac{Y(s)}{U(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \cdots + b_{n-1} s + b_n}{s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n} \quad (3-43)$$



**Figure 3-14**

Block diagram realization of state equation and output equation given by Equations (3-41) and (3-42), respectively.

is given also by Equations (3-41) and (3-42). Figure 3-14 is a block diagram realization of the state equation and output equation given by Equations (3-41) and (3-42), respectively.

There are many ways to obtain state-space representations of systems. Some of them are presented in Problems A-3-4 through A-3-7. Methods for obtaining canonical representations of systems in state space (such as controllable canonical form, observable canonical form, diagonal canonical form, and Jordan canonical form) are presented in Chapter 11.

## 3-6 MECHANICAL SYSTEMS

In this section we shall discuss mathematical modeling of mechanical systems. The fundamental law governing mechanical systems is Newton's second law. It can be applied to any mechanical system. In this section we shall derive mathematical models of two mechanical systems. (Mathematical models of additional mechanical systems will be derived and analyzed throughout the remaining chapters.) Before we discuss mechanical systems, let us review definitions of mass, force, and unit systems.

**Mass.** The *mass* of a body is the quantity of matter in it, which is assumed to be constant. Physically, mass is the property of a body that gives it inertia, that is, resistance to starting and stopping. A body is attracted by Earth, and the magnitude of the force that Earth exerts on it is called its *weight*.

In practical situations, we know the weight  $w$  of a body but not the mass  $m$ . We calculate mass  $m$  from

$$m = \frac{w}{g}$$

where  $g$  is the gravitational acceleration constant. The value of  $g$  varies slightly from point to point on Earth's surface. As a result, the weight of a body varies slightly at different points on Earth's surface, but its mass remains constant. For engineering purposes,  $g$  is taken as

$$g = 9.81 \text{ m/s}^2 = 981 \text{ cm/s}^2 = 32.2 \text{ ft/s}^2 = 386 \text{ in./s}^2$$

Far out in space, a body becomes weightless. Yet its mass remains constant and so the body possesses inertia.

The units of mass are kg, g, lb,  $\text{kg}\cdot\text{s}^2/\text{m}$ , and slug, as shown in Table 3-2. If mass is expressed in units of kilogram (or pound), we call it kilogram mass (or pound mass) to distinguish it from the unit of force, which is termed kilogram force (or pound force). In this book kg is used to denote a kilogram mass and  $\text{kg}_f$  a kilogram force. Similarly, lb denotes a pound mass and  $\text{lb}_f$  a pound force.

A slug is a unit of mass such that, when acted on by 1-pound force, a 1-slug mass accelerates at  $1 \text{ ft/s}^2$  ( $\text{slug} = \text{lb}_f\cdot\text{s}^2/\text{ft}$ ). In other words, if a mass of 1 slug is acted on by 32.2 pounds force, it accelerates at  $32.2 \text{ ft/s}^2 (= g)$ . Hence the mass of a body weighing  $32.2 \text{ lb}_f$  at the earth's surface is 1 slug or

$$m = \frac{w}{g} = \frac{32.2 \text{ lb}_f}{32.2 \text{ ft/s}^2} = 1 \text{ slug}$$

**Table 3–2** Systems of Units

| Quantity \ Systems of units | Absolute systems                                                 |                                                                  |                                                                    | Gravitational systems                           |                                                                    |
|-----------------------------|------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------|--------------------------------------------------------------------|
|                             | Metric                                                           |                                                                  |                                                                    | Metric engineering                              | British engineering                                                |
|                             | SI                                                               | mks                                                              | cgs                                                                |                                                 |                                                                    |
| Length                      | m                                                                | m                                                                | cm                                                                 | m                                               | ft                                                                 |
| Mass                        | kg                                                               | kg                                                               | g                                                                  | $\frac{\text{kg}_f \cdot \text{s}^2}{\text{m}}$ | $\frac{\text{slug} \cdot \text{lb}_f \cdot \text{s}^2}{\text{ft}}$ |
| Time                        | s                                                                | s                                                                | s                                                                  | s                                               | s                                                                  |
| Force                       | $\frac{\text{N}}{= \frac{\text{kg} \cdot \text{m}}{\text{s}^2}}$ | $\frac{\text{N}}{= \frac{\text{kg} \cdot \text{m}}{\text{s}^2}}$ | $\frac{\text{dyn}}{= \frac{\text{g} \cdot \text{cm}}{\text{s}^2}}$ | $\text{kg}_f$                                   | $\text{lb}_f$                                                      |
| Energy                      | $\frac{\text{J}}{= \text{N} \cdot \text{m}}$                     | $\frac{\text{J}}{= \text{N} \cdot \text{m}}$                     | $\frac{\text{erg}}{= \text{dyn} \cdot \text{cm}}$                  | $\text{kg}_f \cdot \text{m}$                    | $\frac{\text{ft} \cdot \text{lb}_f}{\text{or Btu}}$                |
| Power                       | $\frac{\text{W}}{= \frac{\text{N} \cdot \text{m}}{\text{s}}}$    | $\frac{\text{W}}{= \frac{\text{N} \cdot \text{m}}{\text{s}}}$    | $\frac{\text{dyn} \cdot \text{cm}}{\text{s}}$                      | $\frac{\text{kg}_f \cdot \text{m}}{\text{s}}$   | $\frac{\text{ft} \cdot \text{lb}_f}{\text{s}}$<br>or hp            |

**Force.** *Force* can be defined as the cause that tends to produce a change in motion of a body on which it acts. To move a body, force must be applied to it. Two types of forces are capable of acting on a body: *contact* forces and *field* forces. Contact forces are those that come into direct contact with a body, whereas field forces, such as gravitational force and magnetic force, act on a body but do not come into contact with it.

The units of force are newton (N), dyne (dyn),  $\text{kg}_f$ , and  $\text{lb}_f$ . In SI units and the mks system (a metric absolute system) of units the force unit is the newton. The newton is the force that will give a 1-kilogram mass an acceleration of  $1 \text{ m/s}^2$  or

$$1 \text{ N} = 1 \text{ kg} \cdot \text{m/s}^2$$

This means that 9.81 newtons will give a kilogram mass an acceleration of  $9.81 \text{ m/s}^2$ . Since the gravitational acceleration is  $g = 9.81 \text{ m/s}^2$  (as stated earlier, for engineering calculations, the value of  $g$  may be taken as  $9.81 \text{ m/s}^2$  or  $32.2 \text{ ft/s}^2$ ), a mass of 1 kilogram will produce a force on its support of 9.81 newtons.

The force unit in the cgs system (a metric absolute system) is the dyne, which will give a gram mass an acceleration of  $1 \text{ cm/s}^2$  or

$$1 \text{ dyn} = 1 \text{ g} \cdot \text{cm/s}^2$$

The force unit in the metric engineering (gravitational) system is  $\text{kg}_f$ , which is a primary dimension in the system. Similarly, in the British engineering system the force unit is  $\text{lb}_f$ . It is also a primary dimension in this system of units.

**Comments.** SI units for force, mass, and length are the newton (N), kilogram mass (kg), and meter (m). The mks units for force, mass, and length are the same as the SI units. Similarly, the cgs units for force, mass, and length are the dyne (dyn), gram (g), and centimeter (cm), and those for BES units are pound force (lb<sub>f</sub>), slug, and foot (ft). Each of the unit systems is consistent in that the unit of force accelerates the unit of mass 1 unit of length per second per second.

In the systems of units shown in Table 3-2, “s” is used for the second. In engineering papers and books, however, “sec” is commonly used. Therefore, in this book we use “sec”, rather than “s”, for the second.

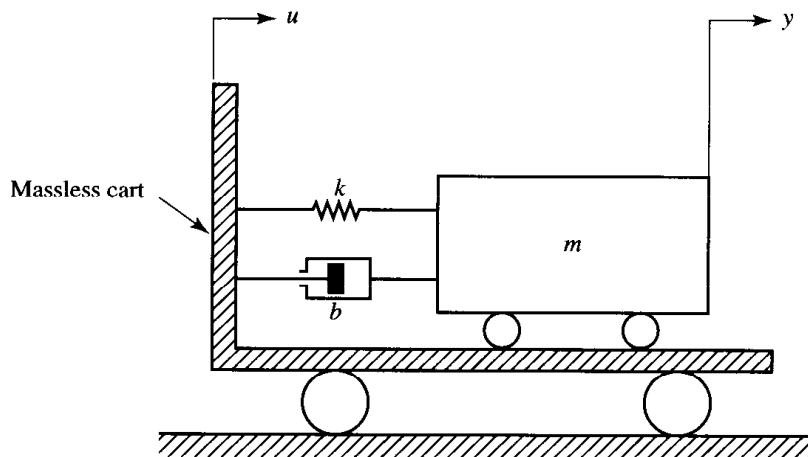
**Mechanical system.** Consider the spring-mass-dashpot system mounted on a massless cart as shown in Figure 3-15. A dashpot is a device that provides viscous friction, or damping. It consists of a piston and oil-filled cylinder. Any relative motion between the piston rod and the cylinder is resisted by the oil because the oil must flow around the piston (or through orifices provided in the piston) from one side of the piston to the other. The dashpot essentially absorbs energy. This absorbed energy is dissipated as heat, and the dashpot does not store any kinetic or potential energy. The dashpot is also called a *damper*.

Let us obtain a mathematical model of this spring-mass-dashpot system mounted on a cart by assuming that the cart is standing still for  $t < 0$ . In this system,  $u(t)$  is the displacement of the cart and is the input to the system. At  $t = 0$ , the cart is moved at a constant speed, or  $\dot{u} = \text{constant}$ . The displacement  $y(t)$  of the mass is the output. (The displacement is relative to the ground.) In this system,  $m$  denotes the mass,  $b$  denotes the viscous friction coefficient, and  $k$  denotes the spring constant. We assume that the friction force of the dashpot is proportional to  $\dot{y} - \dot{u}$  and that the spring is a linear spring; that is, the spring force is proportional to  $y - u$ .

For translational systems, Newton’s second law states that

$$ma = \sum F$$

where  $m$  is a mass,  $a$  is the acceleration of the mass, and  $\sum F$  is the sum of the forces acting on the mass. Applying Newton’s second law to the present system and noting that the cart is massless, we obtain



**Figure 3-15**  
Spring-mass-  
dashpot system  
mounted on a cart.

$$m \frac{d^2 y}{dt^2} = -b \left( \frac{dy}{dt} - \frac{du}{dt} \right) - k(y - u)$$

or

$$m \frac{d^2 y}{dt^2} + b \frac{dy}{dt} + ky = b \frac{du}{dt} + ku \quad (3-44)$$

Equation (3-44) gives a mathematical model of the system considered.

A transfer function model is another way of representing a mathematical model of a linear, time-invariant system. For the present mechanical system, the transfer function model can be obtained as follows: Taking the Laplace transform of each term of Equation (3-44) gives

$$\begin{aligned} \mathcal{L} \left[ m \frac{d^2 y}{dt^2} \right] &= m[s^2 Y(s) - sy(0) - \dot{y}(0)] \\ \mathcal{L} \left[ b \frac{dy}{dt} \right] &= b[sY(s) - y(0)] \\ \mathcal{L}[ky] &= kY(s) \\ \mathcal{L} \left[ b \frac{du}{dt} \right] &= b[sU(s) - u(0)] \\ \mathcal{L}[ku] &= kU(s) \end{aligned}$$

If we set the initial conditions equal to zero, or set  $y(0) = 0$ ,  $\dot{y}(0) = 0$ , and  $u(0) = 0$ , the Laplace transform of Equation (3-44) can be written as

$$(ms^2 + bs + k)Y(s) = (bs + k)U(s)$$

Taking the ratio of  $Y(s)$  to  $U(s)$ , we find the transfer function of the system to be

$$\text{Transfer function} = G(s) = \frac{Y(s)}{U(s)} = \frac{bs + k}{ms^2 + bs + k}$$

Such a transfer-function representation of a mathematical model is used very frequently in control engineering. It should be noted, however, that transfer-function models apply only to linear, time-invariant systems, since the transfer functions are defined only for such systems.

Next we shall obtain a state-space model of this system. We shall first compare the differential equation for this system

$$\ddot{y} + \frac{b}{m} \dot{y} + \frac{k}{m} y = \frac{b}{m} \dot{u} + \frac{k}{m} u$$

with the standard form

$$\ddot{y} + a_1 \dot{y} + a_2 y = b_0 \ddot{u} + b_1 \dot{u} + b_2 u$$

and identify  $a_1$ ,  $a_2$ ,  $b_0$ ,  $b_1$ , and  $b_2$  as follows:

$$a_1 = \frac{b}{m}, \quad a_2 = \frac{k}{m}, \quad b_0 = 0, \quad b_1 = \frac{b}{m}, \quad b_2 = \frac{k}{m}$$

Referring to Equation (3–39), we have

$$\beta_0 = b_0 = 0$$

$$\beta_1 = b_1 - a_1\beta_0 = \frac{b}{m}$$

$$\beta_2 = b_2 - a_1\beta_1 - a_2\beta_0 = \frac{k}{m} - \left(\frac{b}{m}\right)^2$$

Then, referring to Equation (3–38), define

$$x_1 = y - \beta_0 u = y$$

$$x_2 = \dot{x}_1 - \beta_1 u = \dot{x}_1 - \frac{b}{m} u$$

From Equation (3–40) we have

$$\dot{x}_1 = x_2 + \beta_1 u = x_2 + \frac{b}{m} u$$

$$\dot{x}_2 = -a_2 x_1 - a_1 x_2 + \beta_2 u = -\frac{k}{m} x_1 - \frac{b}{m} x_2 + \left[ \frac{k}{m} - \left(\frac{b}{m}\right)^2 \right] u$$

and the output equation becomes

$$y = x_1$$

or

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} \frac{b}{m} \\ \frac{k}{m} - \left(\frac{b}{m}\right)^2 \end{bmatrix} u \quad (3-45)$$

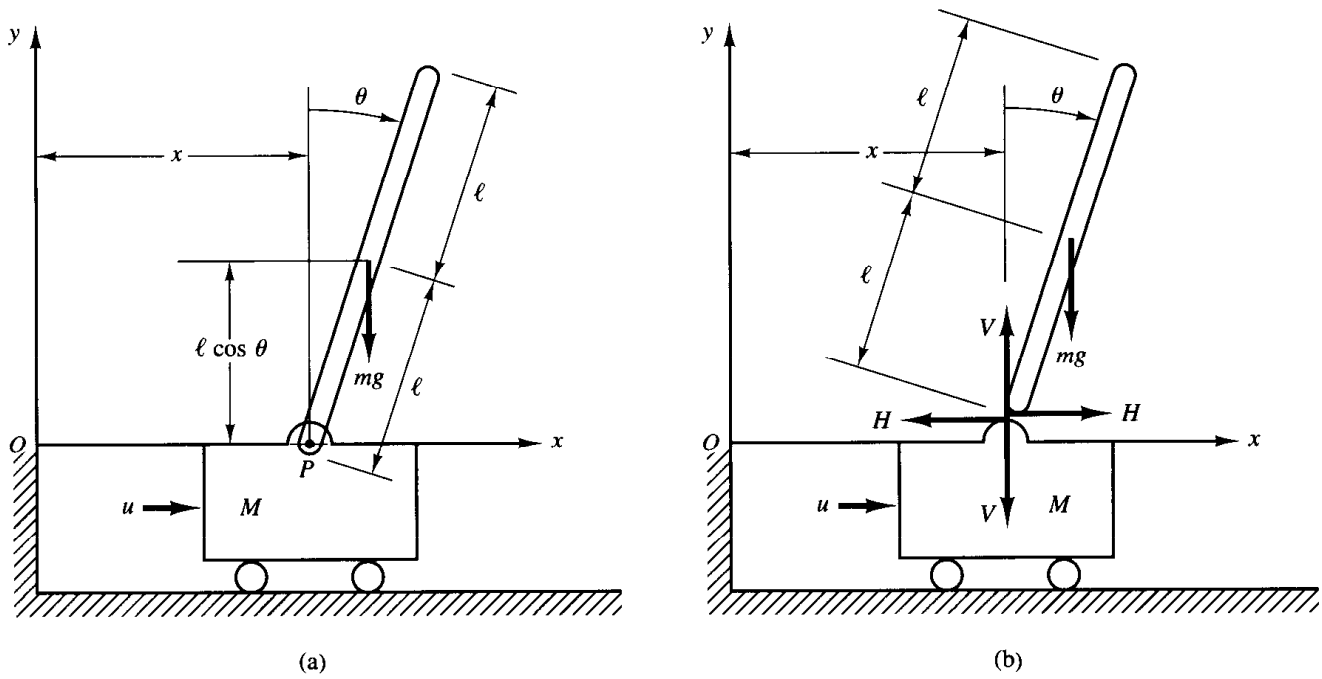
and

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad (3-46)$$

Equations (3–45) and (3–46) give a state-space representation of the system. (Note that this is not the only state-space representation. There are infinitely many state-space representations for the system.)

#### EXAMPLE 3–4

An inverted pendulum mounted on a motor-driven cart is shown in Figure 3–16(a). This is a model of the attitude control of a space booster on takeoff. (The objective of the attitude control problem is to keep the space booster in a vertical position.) The inverted pendulum is unstable in that it may fall over any time in any direction unless a suitable control force is applied. Here we consider only a two-dimensional problem in which the pendulum moves only in the plane of the page. The control force  $u$  is applied to the cart. Assume that the center of gravity of the pendulum rod is at its geometric center. Obtain a mathematical model for the system. Assume that the mass  $m$  of the pendulum rod is 0.1 kg, the mass  $M$  of the cart is 2 kg, and the length  $2l$  of the pendulum rod is 1 m, or



**Figure 3-16**  
(a) Inverted pendulum system; (b) free-body diagram.

$$m = 0.1 \text{ kg}, \quad M = 2 \text{ kg}, \quad 2l = 1 \text{ m}$$

Define the angle of the rod from the vertical line as  $\theta$ . Define also the  $(x, y)$  coordinates of the center of gravity of the pendulum rod as  $(x_G, y_G)$ . Then

$$x_G = x + l \sin \theta$$

$$y_G = l \cos \theta$$

To derive the equations of motion for the system, consider the free-body diagram shown in Figure 3-16(b). The rotational motion of the pendulum rod about its center of gravity can be described by

$$I\ddot{\theta} = Vl \sin \theta - Hl \cos \theta \quad (3-47)$$

where  $I$  is the moment of inertia of the rod about its center of gravity.

The horizontal motion of center of gravity of pendulum rod is given by

$$m \frac{d^2}{dt^2} (x + l \sin \theta) = H \quad (3-48)$$

The vertical motion of center of gravity of pendulum rod is

$$m \frac{d^2}{dt^2} (l \cos \theta) = V - mg \quad (3-49)$$

The horizontal motion of cart is described by

$$M \frac{d^2 x}{dt^2} = u - H \quad (3-50)$$

Equations (3–47) through (3–50) describe the motion of the inverted-pendulum-on-the-cart system. Because these equations involve  $\sin \theta$  and  $\cos \theta$ , they are nonlinear equations.

If we assume angle  $\theta$  to be small, Equations (3–47) through (3–50) may be linearized as follows:

$$I\ddot{\theta} = V l \theta - H l \quad (3-51)$$

$$m(\ddot{x} + l\ddot{\theta}) = H \quad (3-52)$$

$$0 = V - mg \quad (3-53)$$

$$M\ddot{x} = u - H \quad (3-54)$$

From Equations (3–52) and (3–54), we obtain

$$(M + m)\ddot{x} + ml\ddot{\theta} = u \quad (3-55)$$

From Equations (3–51) and (3–53), we have

$$\begin{aligned} I\ddot{\theta} &= mgl\theta - Hl \\ &= mgl\theta - l(m\ddot{x} + ml\ddot{\theta}) \end{aligned}$$

or

$$(I + ml^2)\ddot{\theta} + ml\ddot{x} = mgl\theta \quad (3-56)$$

Equations (3–55) and (3–56) describe the motion of the inverted-pendulum-on-the-cart system. They constitute a mathematical model of the system. (Later in Chapters 12 and 13, we design controllers to keep the pendulum upright in the presence of disturbances.)

### 3-7 ELECTRICAL SYSTEMS

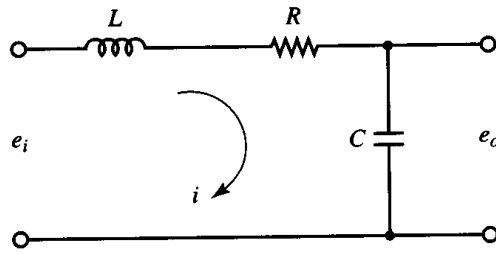
In this section we shall deal with electrical circuits involving resistors, capacitors, and inductors.

Basic laws governing electrical circuits are Kirchhoff's current law and voltage law. Kirchhoff's current law (node law) states that the algebraic sum of all currents entering and leaving a node is zero. (This law can also be stated as follows: The sum of currents entering a node is equal to the sum of currents leaving the same node.) Kirchhoff's voltage law (loop law) states that at any given instant the algebraic sum of the voltages around any loop in an electrical circuit is zero. (This law can also be stated as follows: The sum of the voltage drops is equal to the sum of the voltage rises around a loop.) A mathematical model of an electrical circuit can be obtained by applying one or both of Kirchhoff's laws to it.

This section deals with simple electrical circuits. Mathematical modeling of operational amplifier systems is presented in Chapter 5.

**LRC circuit.** Consider the electrical circuit shown in Figure 3–17. The circuit consists of an inductance  $L$  (henry), a resistance  $R$  (ohm), and a capacitance  $C$  (farad). Applying Kirchhoff's voltage law to the system, we obtain the following equations:





**Figure 3-17**  
Electrical circuit.

$$L \frac{di}{dt} + Ri + \frac{1}{C} \int i dt = e_i \quad (3-57)$$

$$\frac{1}{C} \int i dt = e_o \quad (3-58)$$

Equations (3-57) and (3-58) give a mathematical model of the circuit.

A transfer function model of the circuit can also be obtained as follows: Taking the Laplace transforms of Equations (3-57) and (3-58), assuming zero initial conditions, we obtain

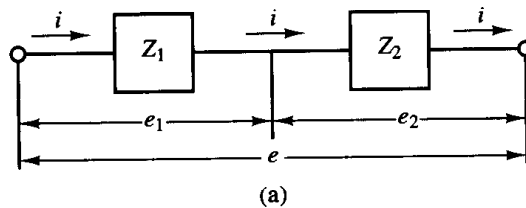
$$LsI(s) + RI(s) + \frac{1}{C} \frac{1}{s} I(s) = E_i(s)$$

$$\frac{1}{C} \frac{1}{s} I(s) = E_o(s)$$

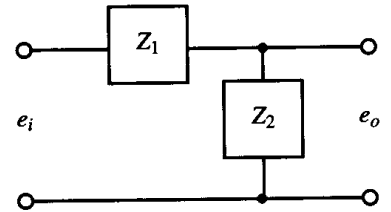
If  $e_i$  is assumed to be the input and  $e_o$  the output, then the transfer function of this system is found to be

$$\frac{E_o(s)}{E_i(s)} = \frac{1}{LCs^2 + RCs + 1} \quad (3-59)$$

**Complex impedances.** In deriving transfer functions for electrical circuits, we frequently find it convenient to write the Laplace-transformed equations directly, without writing the differential equations. Consider the system shown in Figure 3-18(a). In this system,  $Z_1$  and  $Z_2$  represent complex impedances. The complex impedance  $Z(s)$  of a two-terminal circuit is the ratio of  $E(s)$ , the Laplace transform of the voltage across the terminals, to  $I(s)$ , the Laplace transform of the current through the element, under the assumption that the initial conditions are zero, so that  $Z(s) = E(s)/I(s)$ . If the two-terminal elements is a resistance  $R$ , capacitance  $C$ , or inductance  $L$ , then the complex impedance is given by  $R$ ,  $1/Cs$ , or  $Ls$ , respectively. If complex impedances are connected in series, the total impedance is the sum of the individual complex impedances.



(a)



(b)

**Figure 3-18**  
Electrical circuits.

Remember that the impedance approach is valid only if the initial conditions involved are all zeros. Since the transfer function requires zero initial conditions, the impedance approach can be applied to obtain the transfer function of the electrical circuit. This approach greatly simplifies the derivation of transfer functions of electrical circuits.

Consider the circuit shown in Figure 3–18(b). Assume that the voltages  $e_i$  and  $e_o$  are the input and output of the circuit, respectively. Then the transfer function of this circuit is

$$\frac{E_o(s)}{E_i(s)} = \frac{Z_2(s)}{Z_1(s) + Z_2(s)}$$

For the system shown in Figure 3–17,

$$Z_1 = Ls + R, \quad Z_2 = \frac{1}{Cs}$$

Hence the transfer function  $E_o(s)/E_i(s)$  can be found as follows:

$$\frac{E_o(s)}{E_i(s)} = \frac{\frac{1}{Cs}}{Ls + R + \frac{1}{Cs}} = \frac{1}{LCs^2 + RCs + 1}$$

which is, of course, identical to Equation (3–59).

**State-space representation.** A state-space model of the system shown in Figure 3–17 may be obtained as follows: First, note that the differential equation for the system can be obtained from Equation (3–59) as

$$\ddot{e}_o + \frac{R}{L} \dot{e}_o + \frac{1}{LC} e_o = \frac{1}{LC} e_i$$

Then by defining state variables by

$$x_1 = e_o$$

$$x_2 = \dot{e}_o$$

and the input and output variables by

$$u = e_i$$

$$y = e_o = x_1$$

we obtain

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{1}{LC} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{LC} \end{bmatrix} u$$

and

$$y = [1 \quad 0] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

These two equations give a mathematical model of the system in state space.

**Transfer functions of cascaded elements.** Many feedback systems have components that load each other. Consider the system shown in Figure 3-19. Assume that  $e_i$  is the input and  $e_o$  is the output. In this system the second stage of the circuit ( $R_2C_2$  portion) produces a loading effect on the first stage ( $R_1C_1$  portion). The equations for this system are

$$\frac{1}{C_1} \int (i_1 - i_2) dt + R_1 i_1 = e_i \quad (3-60)$$

and

$$\frac{1}{C_1} \int (i_2 - i_1) dt + R_2 i_2 + \frac{1}{C_2} \int i_2 dt = 0 \quad (3-61)$$

$$\frac{1}{C_2} \int i_2 dt = e_o \quad (3-62)$$

Taking the Laplace transforms of Equations (3-60) through (3-62), respectively, assuming zero initial conditions, we obtain

$$\frac{1}{C_1 s} [I_1(s) - I_2(s)] + R_1 I_1(s) = E_i(s) \quad (3-63)$$

$$\frac{1}{C_1 s} [I_2(s) - I_1(s)] + R_2 I_2(s) + \frac{1}{C_2 s} I_2(s) = 0 \quad (3-64)$$

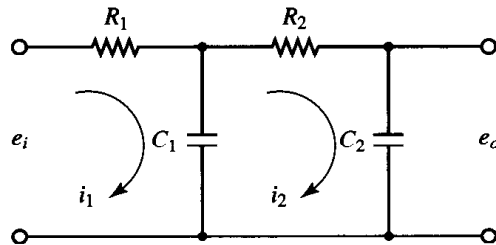
$$\frac{1}{C_2 s} I_2(s) = E_o(s) \quad (3-65)$$

Eliminating  $I_1(s)$  from Equations (3-63) and (3-64) and writing  $E_i(s)$  in terms of  $I_2(s)$ , we find the transfer function between  $E_o(s)$  and  $E_i(s)$  to be

$$\begin{aligned} \frac{E_o(s)}{E_i(s)} &= \frac{1}{(R_1 C_1 s + 1)(R_2 C_2 s + 1) + R_1 C_2 s} \\ &= \frac{1}{R_1 C_1 R_2 C_2 s^2 + (R_1 C_1 + R_2 C_2 + R_1 C_2) s + 1} \end{aligned} \quad (3-66)$$

The term  $R_1 C_2 s$  in the denominator of the transfer function represents the interaction of two simple  $RC$  circuits. Since  $(R_1 C_1 + R_2 C_2 + R_1 C_2)^2 > 4 R_1 C_1 R_2 C_2$ , the two roots of the denominator of Equation (3-66) are real.

The present analysis shows that, if two  $RC$  circuits are connected in cascade so that the output from the first circuit is the input to the second, the overall transfer function



**Figure 3-19**  
Electrical system.